ELECTRIC FIELDS

18.1 Electric fields and field lines

Candidates should be able to:

1 understand that an electric field is an example of a field of force and define electric field as force per unit positive charge

2 recall and use $F = qE$ for the force on a charge in an electric field

3 represent an electric field by means of field lines

18.2 Uniform electric fields

Candidates should be able to:

1 recall and use $E = \Delta V / \Delta d$ to calculate the field strength of the uniform field between charged parallel plates

2 describe the effect of a uniform electric field on the motion of charged particles

18.3 Electric force between point charges

Candidates should be able to:

1 understand that, for a point outside a spherical conductor, the charge on the sphere may be considered to be a point charge at its centre

2 recall and use Coulomb's law $F = Q_1Q_2 / (4\pi\varepsilon_0 r^2)$ for the force between two point charges in free space

18.4 Electric field of a point charge

Candidates should be able to:

1 recall and use $E = Q / (4\pi\varepsilon_0 r^2)$ for the electric field strength due to a point charge in free space

18.5 Electric potential

Candidates should be able to:

1 define electric potential at a point as the work done per unit positive charge in bringing a small test charge from infinity to the point

2 recall and use the fact that the electric field at a point is equal to the negative of potential gradient at that point

3 use $V = Q / (4\pi\varepsilon_0 r)$ for the electric potential in the field due to a point charge

4 understand how the concept of electric potential leads to the electric potential energy of two point charges and use $E_P = Qq / (4\pi\varepsilon_0 r)$

1. M/J/07/Q3

Two charged points A and B are separated by a distance of 6.0 cm, as shown in Fig. 3.1.

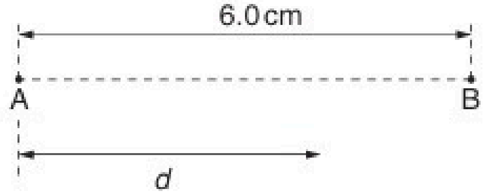


Fig. 3.1

The variation with distance d from A of the electric field strength E along the line AB is shown in Fig. 3.2.

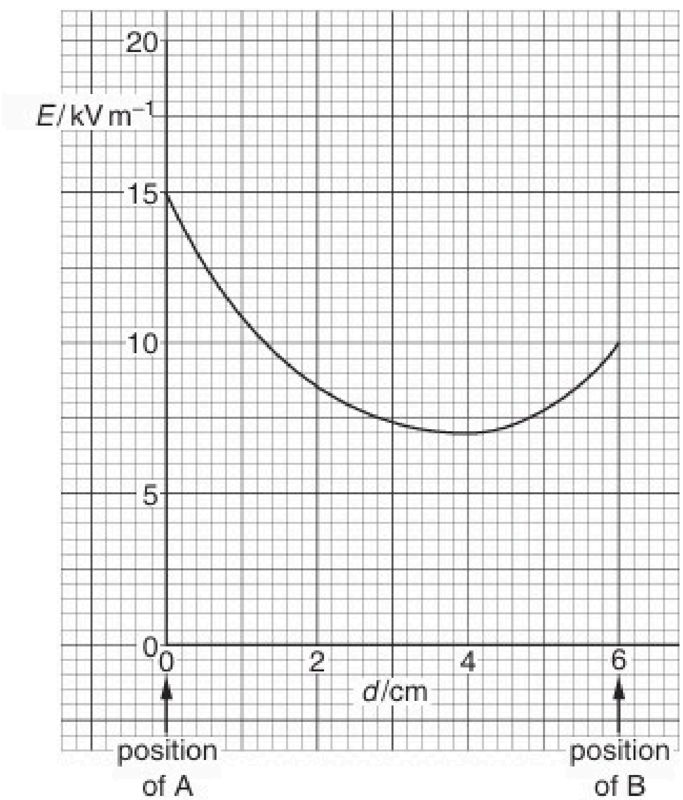


Fig. 3.2

An electron is emitted with negligible speed from A and travels along AB.

(a) State the relation between electric field strength E and potential V.

....

.... [2]

(b) The area below the line of the graph of Fig. 3.2 represents the potential difference between A and B.

Use Fig. 3.2 to determine the potential difference between A and B.

potential difference = ... V [4]

(c) Use your answer to **(b)** to calculate the speed of the electron as it reaches point B.

speed = ... $\mathrm{m\,s^{-1}}$ [2]

(d) **(i)** Use Fig. 3.2 to determine the value of *d* at which the electron has maximum acceleration.

d = ... cm [1]

(ii) Without any further calculation, describe the variation with distance *d* of the acceleration of the electron.

....

....

.... [2]

2. O/N/07/Q4

A small charged metal sphere is situated in an earthed metal box. Fig. 4.1 illustrates the electric field between the sphere and the metal box.

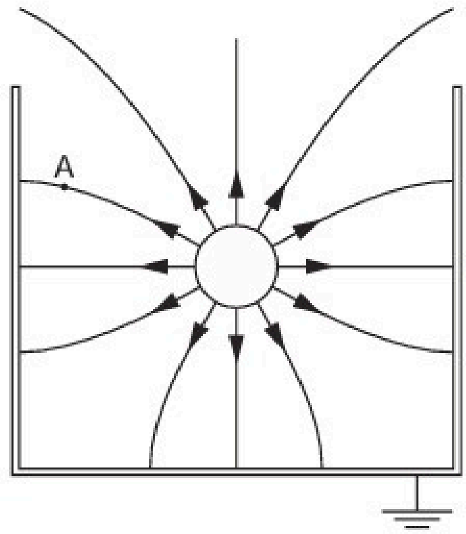


Fig. 4.1

(a) By reference to Fig. 4.1, state and explain

(i) whether the sphere is positively or negatively charged,

....

....

....[2]

(ii) why it appears as if the charge on the sphere is concentrated at the centre of the sphere.

....

....[1]

(b) On Fig. 4.1, draw an arrow to show the direction of the force on a stationary electron situated at point A. [2]

(c) The radius r of the sphere is 2.4 cm. The magnitude of the charge q on the sphere is 0.76 nC.

(i) Use the expression

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

to calculate a value for the magnitude of the potential V at the surface of the sphere.

V = ...V [2]

(ii) State the sign of the charge induced on the inside of the metal box. Hence explain whether the actual magnitude of the potential will be greater or smaller than the value calculated in **(i)**.

....

....

....

....[3]

(d) A lead sphere is placed in a lead box in free space, in a similar arrangement to that shown in Fig. 4.1. Explain why it is **not** possible for the gravitational field to have a similar shape to that of the electric field.

....

....

....[1]

3. M/J/08/Q4

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) Two isolated point charges A and B are separated by a distance of 30.0 cm, as shown in Fig. 4.1.

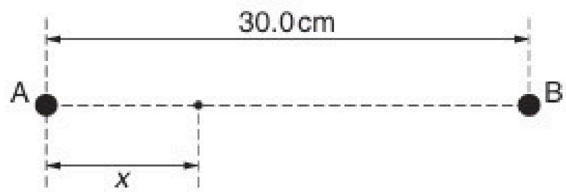


Fig. 4.1

The charge at A is + 3.6×10^{-9} C.

The variation with distance x from A along AB of the potential V is shown in Fig. 4.2.

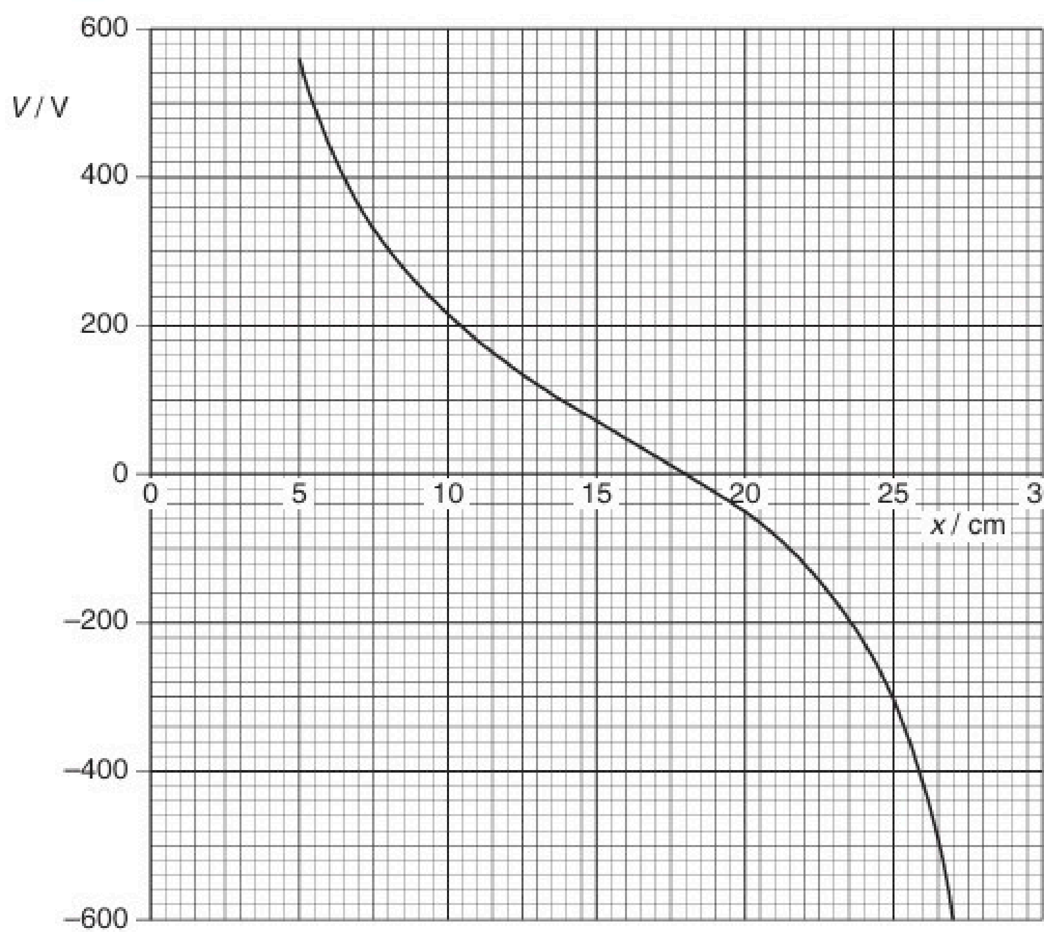


Fig. 4.2

(i) State the value of x at which the potential is zero.

x = ... cm [1]

(ii) Use your answer in **(i)** to determine the charge at B.

charge = ... C [3]

(c) A small test charge is now moved along the line AB in **(b)** from $x = 5.0\,\text{cm}$ to $x = 27\,\text{cm}$. State and explain the value of x at which the force on the test charge will be maximum.

....

....

....

....[3]

4. O/N/09/41/Q5

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) An α-particle is emitted from a radioactive source with kinetic energy of 4.8 MeV.

The α-particle travels in a vacuum directly towards a gold ($^{197}_{79}Au$) nucleus, as illustrated in Fig. 5.1.

Fig. 5.1

The α-particle and the gold nucleus may be considered to be point charges in an isolated system.

(i) Explain why, as the α-particle approaches the gold nucleus, it comes to rest.

....

....

.... [2]

(ii) For the closest approach of the α-particle to the gold nucleus determine

1. their separation,

separation = ... m [3]

2. the magnitude of the force on the α-particle.

force = ... N [2]

5. M/J/10/41/Q4

(a) Explain what is meant by the *potential energy* of a body.

....

....

....[2]

(b) Two deuterium (^2_1H) nuclei each have initial kinetic energy E_K and are initially separated by a large distance.
The nuclei may be considered to be spheres of diameter 3.8×10^{-15} m with their masses and charges concentrated at their centres.
The nuclei move from their initial positions to their final position of just touching, as illustrated in Fig. 4.1.

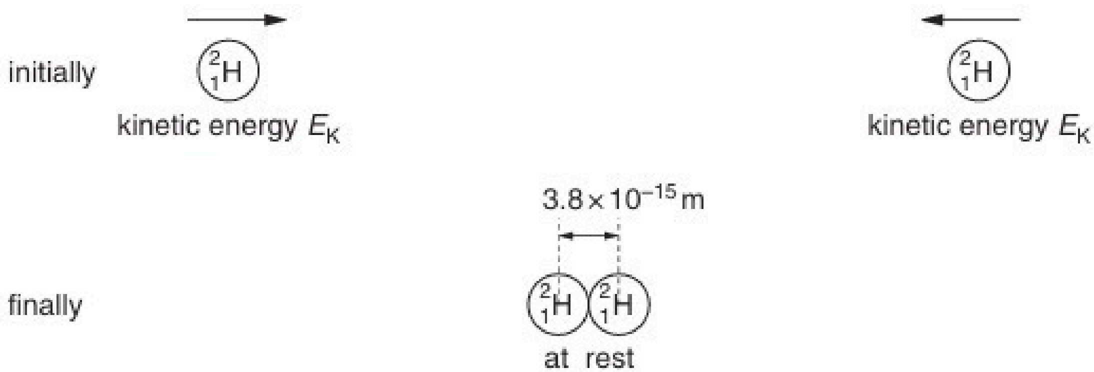


Fig. 4.1

(i) For the two nuclei approaching each other, calculate the total change in

1. gravitational potential energy,

energy = ... J [3]

2. electric potential energy.

energy = ... J [3]

(ii) Use your answers in **(i)** to show that the initial kinetic energy E_K of each nucleus is 0.19 MeV.

[2]

(iii) The two nuclei may rebound from each other. Suggest one other effect that could happen to the two nuclei if the initial kinetic energy of each nucleus is greater than that calculated in **(ii)**.

....

....[1]

6. M/J/10/42/Q4

Two point charges A and B each have a charge of $+6.4 \times 10^{-19}$ C. They are separated in a vacuum by a distance of 12.0 μm, as shown in Fig. 4.1.

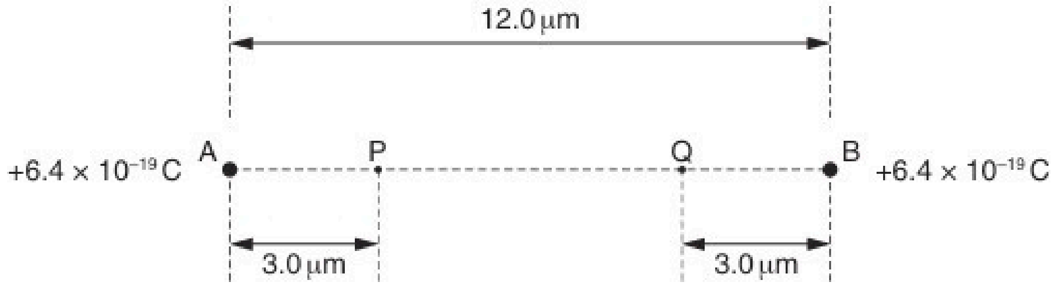


Fig. 4.1

Points P and Q are situated on the line AB. Point P is 3.0 μm from charge A and point Q is 3.0 μm from charge B.

(a) Calculate the force of repulsion between the charges A and B.

force = ... N [3]

(b) Explain why, without any calculation, when a small test charge is moved from point P to point Q, the net work done is zero.

....

....

.... [2]

(c) Calculate the work done by an electron in moving from the midpoint of line AB to point P.

work done = ... J [4]

7. M/J/11/41/Q4

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) Two small spherical charged particles P and Q may be assumed to be point charges located at their centres. The particles are in a vacuum.

Particle P is fixed in position. Particle Q is moved along the line joining the two charges, as illustrated in Fig. 4.1.

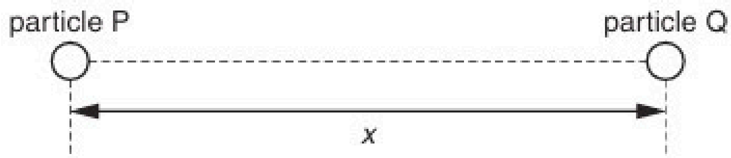


Fig. 4.1

The variation with separation x of the electric potential energy E_P of particle Q is shown in Fig. 4.2.

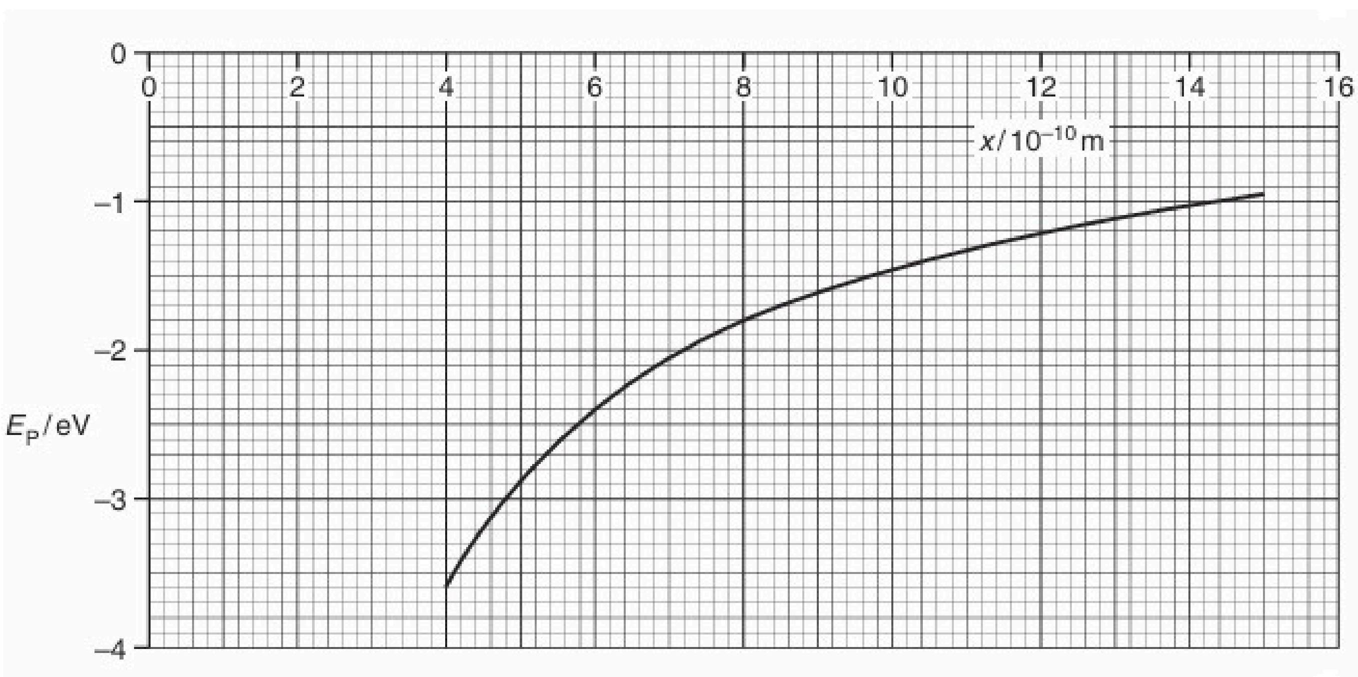


Fig. 4.2

(i) State how the magnitude of the electric field strength is related to potential gradient.

....

.... [1]

(ii) Use your answer in **(i)** to show that the force on particle Q is proportional to the gradient of the curve of Fig. 4.2.

....

....

....[2]

(c) The magnitude of the charge on each of the particles P and Q is 1.6×10^{-19} C.
Calculate the separation of the particles at the point where particle Q has electric potential energy equal to –5.1 eV.

separation = ... m [4]

(d) By reference to Fig. 4.2, state and explain

(i) whether the two charges have the same, or opposite, sign,

....

....

....[2]

(ii) the effect, if any, on the shape of the graph of doubling the charge on particle P.

....

....

....[2]

8. M/J/11/42/Q1

(a) State what is meant by a *field of force*.

....

.... [1]

(b) Gravitational fields and electric fields are two examples of fields of force.
State one similarity and one difference between these two fields of force.

similarity: ....

....

difference: ....

....

....
[3]

(c) Two protons are isolated in space. Their centres are separated by a distance R.
Each proton may be considered to be a point mass with point charge.
Determine the magnitude of the ratio

$$\frac{\text{force between protons due to electric field}}{\text{force between protons due to gravitational field}}.$$

ratio = ... [3]

9. O/N/11/41/Q4

Two small charged metal spheres A and B are situated in a vacuum. The distance between the centres of the spheres is 12.0 cm, as shown in Fig. 4.1.

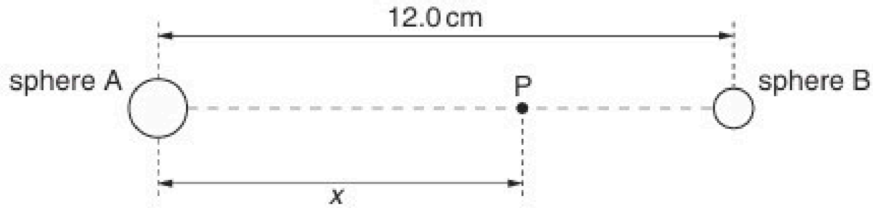


Fig. 4.1 (not to scale)

The charge on each sphere may be assumed to be a point charge at the centre of the sphere.
Point P is a movable point that lies on the line joining the centres of the spheres and is distance x from the centre of sphere A.
The variation with distance x of the electric field strength E at point P is shown in Fig. 4.2.

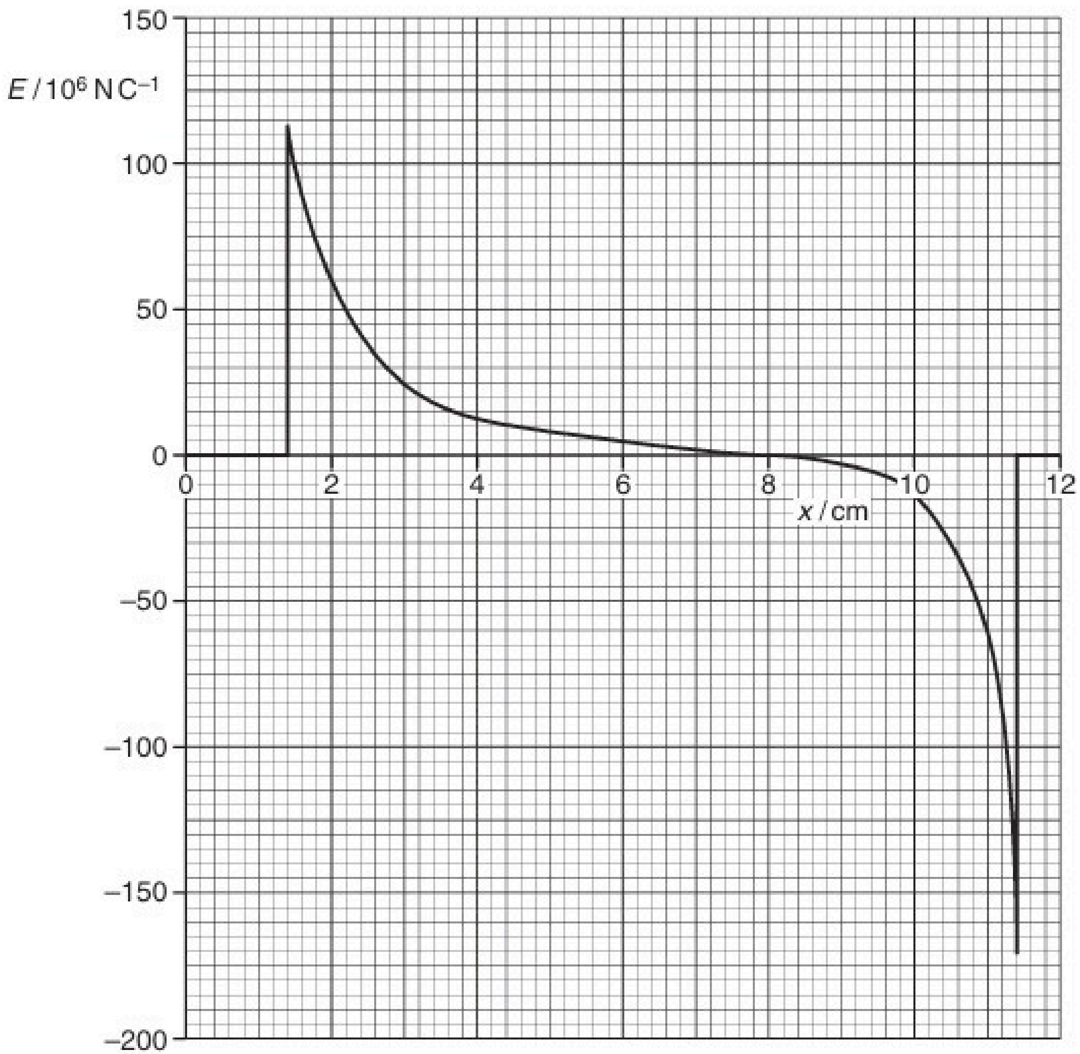


Fig. 4.2

(a) State the evidence provided by Fig. 4.2 for the statements that

(i) the spheres are conductors,

....

....[1]

(ii) the charges on the spheres are either both positive or both negative.

....

....

....[2]

(b) **(i)** State the relation between electric field strength E and potential gradient at a point.

....

....[1]

(ii) Use Fig. 4.2 to state and explain the distance x at which the rate of change of potential with distance is

1. maximum,

....

....

....[2]

2. minimum.

....

....

....[2]

10. M/J/12/41/Q5

(a) Define *electric field strength.*

....

.... [1]

(b) An isolated metal sphere is to be used to store charge at high potential. The charge stored may be assumed to be a point charge at the centre of the sphere. The sphere has a radius of 25 cm. Electrical breakdown (a spark) occurs in the air surrounding the sphere when the electric field strength at the surface of the sphere exceeds $1.8 \times 10^{4}\,\mathrm{V\,cm^{-1}}$.

(i) Show that the maximum charge that can be stored on the sphere is $12.5\,\mu\mathrm{C}$.

[2]

(ii) Calculate the potential of the sphere for this maximum charge.

potential = ... V [2]

11. M/J/12/42/Q4

A charged point mass is situated in a vacuum. A proton travels directly towards the mass, as illustrated in Fig. 4.1.

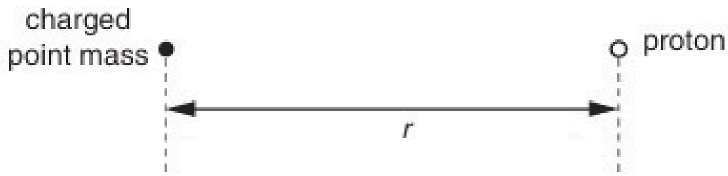


Fig. 4.1

When the separation of the mass and the proton is r, the electric potential energy of the system is U_P.

The variation with r of the potential energy U_P is shown in Fig. 4.2.

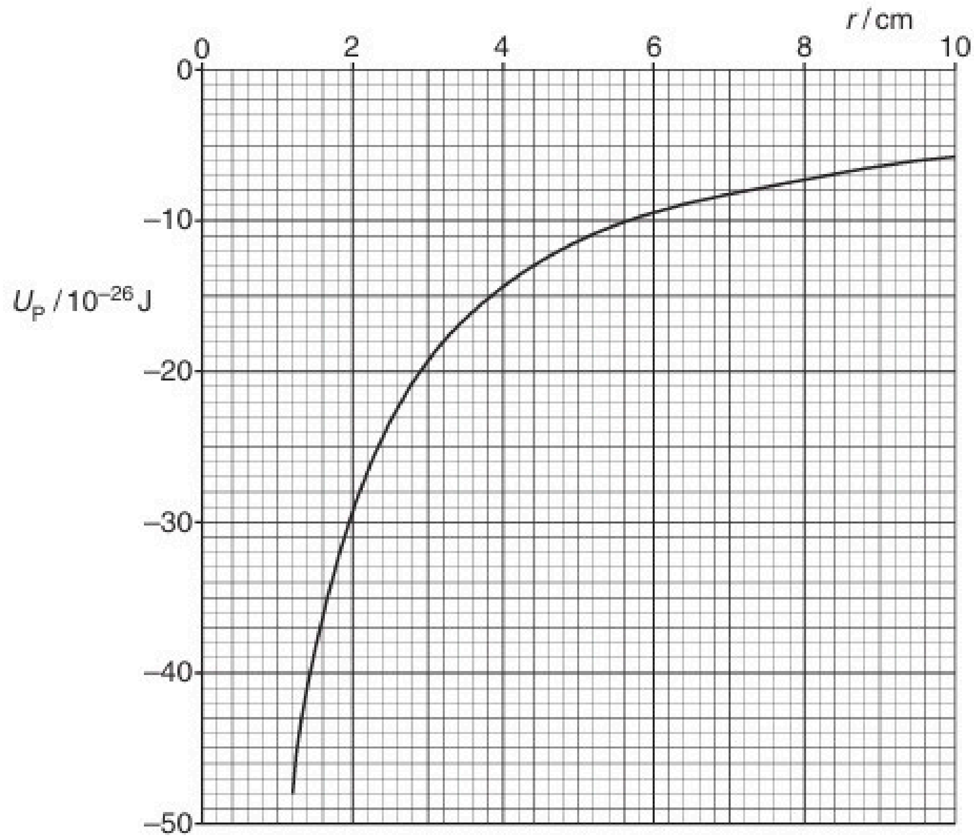


Fig. 4.2

(a) (i) Use Fig. 4.2 to state and explain whether the mass is charged positively or negatively.

....

....

.... [2]

(ii) The gradient at a point on the graph of Fig. 4.2 is G.
Show that the electric field strength E at this point due to the charged point mass is given by the expression

$$Eq = G$$

where q is the charge at this point.

....

....

.... [2]

(b) Use the expression in **(a)(ii)** and Fig. 4.2 to determine the electric field strength at a distance of 4.0 cm from the charged point mass.

field strength = ... $\mathrm{V\,m^{-1}}$ [4]

12. O/N/12/43/Q3

(a) State what is meant by a line of force in

(i) a gravitational field,

....

.... [1]

(ii) an electric field.

....

.... [2]

(b) A charged metal sphere is isolated in space.
State one similarity and one difference between the gravitational force field and the electric force field around the sphere.

similarity: ....

....

difference: ....

....

....
[3]

(c) Two horizontal metal plates are separated by a distance of 1.8 cm in a vacuum.
A potential difference of 270 V is maintained between the plates, as shown in Fig. 3.1.

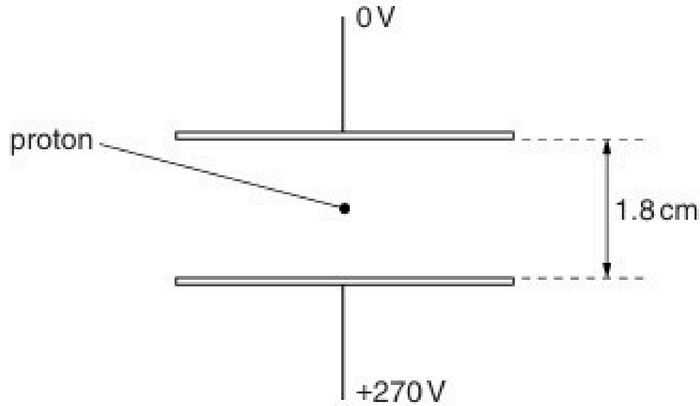


Fig. 3.1

A proton is in the space between the plates.
Explain quantitatively why, when predicting the motion of the proton between the plates, the gravitational field is not taken into consideration.

[4]

13. M/J/13/41/Q4

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) A charged particle is accelerated from rest in a vacuum through a potential difference V. Show that the final speed v of the particle is given by the expression

$$v = \sqrt{\left(\frac{2Vq}{m}\right)}$$

where $\frac{q}{m}$ is the ratio of the charge to the mass (the specific charge) of the particle.

[2]

(c) A particle with specific charge $+9.58 \times 10^{7}\,\mathrm{C\,kg^{-1}}$ is moving in a vacuum towards a fixed metal sphere, as illustrated in Fig. 4.1.

Fig. 4.1

The initial speed of the particle is $2.5 \times 10^{5}\,\mathrm{m\,s^{-1}}$ when it is a long distance from the sphere.
The sphere is positively charged and has a potential of +470 V.

Use the expression in **(b)** to determine whether the particle will reach the surface of the sphere.

14. O/N/13/41/Q4

An α-particle and a proton are at rest a distance 20 μm apart in a vacuum, as illustrated in Fig. 4.1.

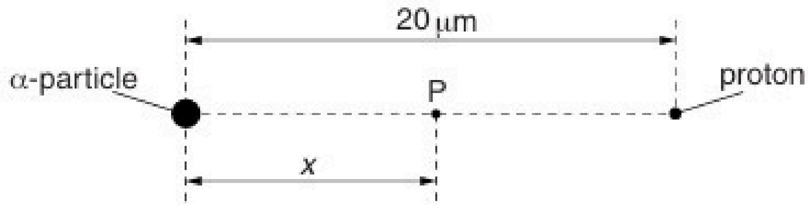


Fig. 4.1

(a) **(i)** State Coulomb's law.

....

....

.... [2]

(ii) The α-particle and the proton may be considered to be point charges. Calculate the electric force between the α-particle and the proton.

force = ... N [2]

(b) **(i)** Define *electric field strength*.

....

....

.... [2]

(ii) A point P is distance x from the α-particle along the line joining the α-particle to the proton (see Fig. 4.1). The variation with distance x of the electric field strength E_α due to the α-particle alone is shown in Fig. 4.2.

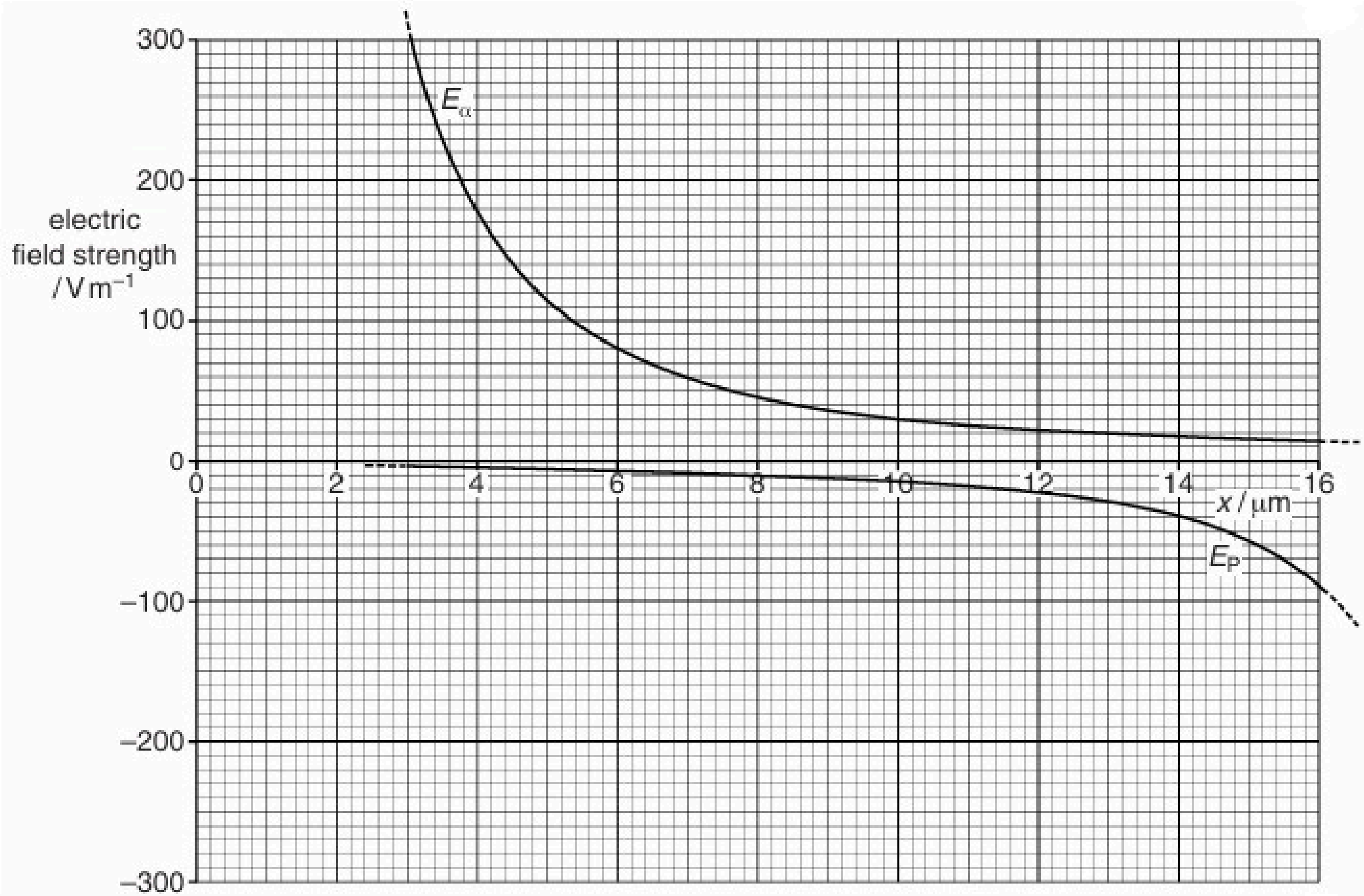


Fig. 4.2

The variation with distance x of the electric field strength E_P due to the proton alone is also shown in Fig. 4.2.

1. Explain why the two separate electric fields have opposite signs.

....

....

....[2]

2. On Fig. 4.2, sketch the variation with x of the combined electric field due to the α-particle and the proton for values of x from 4 μm to 16 μm. [3]

15. O/N/13/43/Q3

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) Two point charges A and B are separated by a distance of 20 nm in a vacuum, as illustrated in Fig. 3.1.

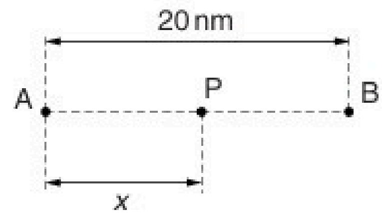


Fig. 3.1

A point P is a distance x from A along the line AB.
The variation with distance x of the electric potential V_A due to charge A alone is shown in Fig. 3.2.

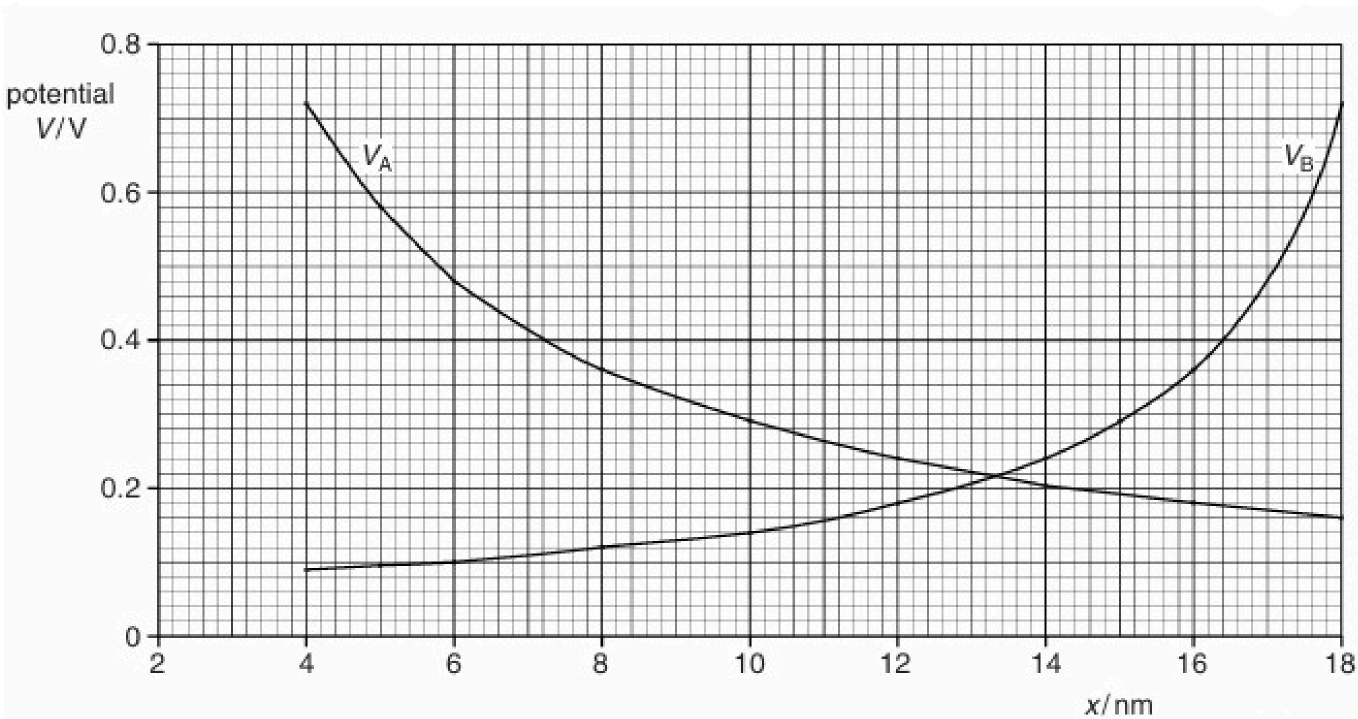


Fig. 3.2

The variation with distance x of the electric potential V_B due to charge B alone is also shown in Fig. 3.2.

(i) State and explain whether the charges A and B are of the same, or opposite, sign.

....

....

.... [2]

(ii) By reference to Fig. 3.2, state how the combined electric potential due to both charges may be determined.

....

.... [1]

(iii) Without any calculation, use Fig. 3.2 to estimate the distance x at which the combined electric potential of the two charges is a minimum.

x = ... nm [1]

(iv) The point P is a distance $x = 10\,\text{nm}$ from A.
An α-particle has kinetic energy E_K when at infinity.

Use Fig. 3.2 to determine the minimum value of E_K such that the α-particle may travel from infinity to point P.

E_K = ... J [3]

16. M/J/14/41/Q5

An isolated solid metal sphere of radius r is given a positive charge. The distance from the centre of the sphere is x.

(a) The electric potential at the surface of the sphere is V_0.

On the axes of Fig. 5.1, sketch a graph to show the variation with distance x of the electric potential due to the charged sphere, for values of x from $x = 0$ to $x = 4r$.

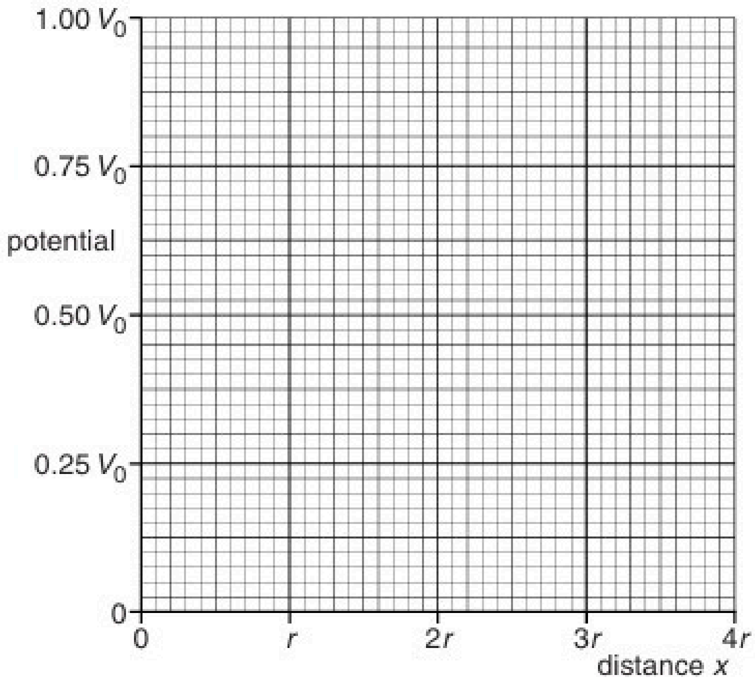


Fig. 5.1

[3]

(b) The electric field strength at the surface of the sphere is E_0.

On the axes of Fig. 5.2, sketch a graph to show the variation with distance x of the electric field strength due to the charged sphere, for values of x from $x = 0$ to $x = 4r$.

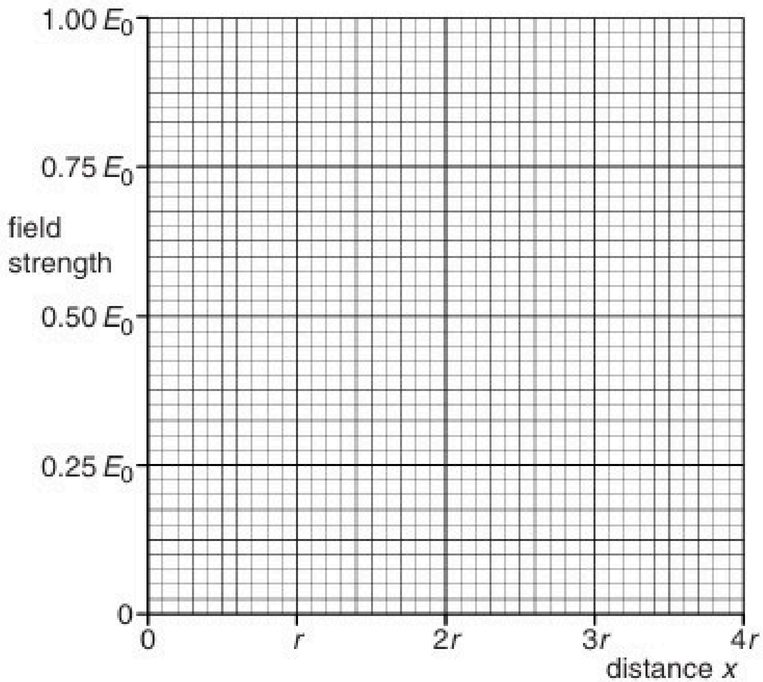


Fig. 5.2

[3]

17. M/J/14/42/Q4

A helium nucleus contains two protons.

In a model of the helium nucleus, each proton is considered to be a charged point mass. The separation of these point masses is assumed to be 2.0×10^{-15} m.

(a) For the two protons in this model, calculate

(i) the electrostatic force,

electrostatic force = ... N [2]

(ii) the gravitational force.

gravitational force = ... N [2]

(b) Using your answers in **(a)**, suggest why

(i) there must be some other force between the protons in the nucleus,

....

....

....

.... [3]

(ii) this additional force must have a short range.

....

....

.... [2]

18. O/N/14/41/Q5

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) An isolated solid metal sphere is positively charged.

The variation of the potential V with distance x from the centre of the sphere is shown in Fig. 5.1.

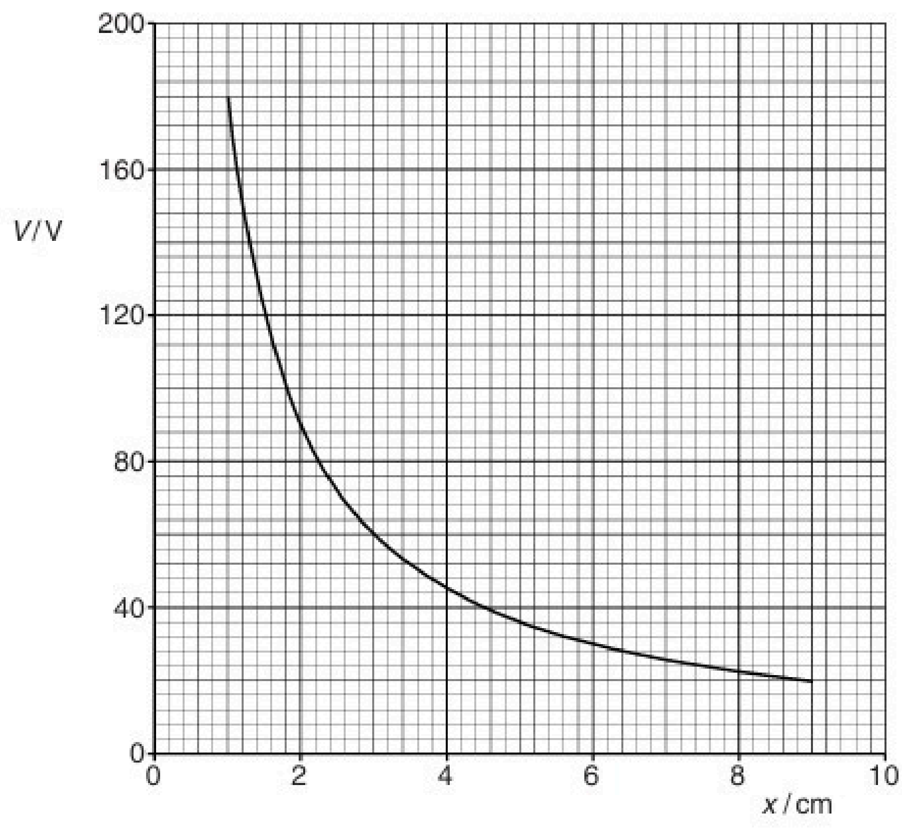


Fig. 5.1

Use Fig. 5.1 to suggest

(i) why the radius of the sphere cannot be greater than 1.0 cm,

....

.... [1]

(ii) that the charge on the sphere behaves as if it were a point charge.

[3]

(c) Assuming that the charge on the sphere does behave as a point charge, use data from Fig. 5.1 to determine the charge on the sphere.

charge = ... C [2]

19. O/N/14/43/Q5

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) An isolated metal sphere is charged to a potential V. The charge on the sphere is q.
The charge on the sphere may be considered to act as a point charge at the centre of the sphere.

The variation with potential V of the charge q on the sphere is shown in Fig. 5.1.

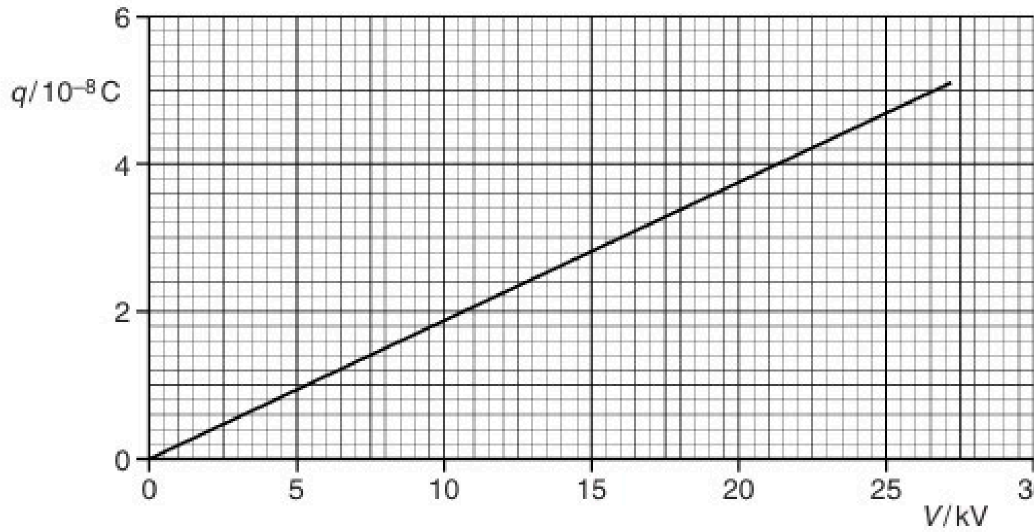


Fig. 5.1

Use Fig. 5.1 to determine

(i) the radius of the sphere,

radius = ... m [2]

(ii) the energy required to increase the potential of the sphere from zero to 24 kV.

energy = ... J [3]

(c) The sphere in **(b)** discharges by causing sparks when the electric field strength at the surface of the sphere is greater than $2.0 \times 10^6 \text{ V m}^{-1}$.

Use your answer in **(b)(i)** to calculate the maximum potential to which the sphere can be charged.

potential = ... V [3]

20. M/J/15/41/Q5

A charged metal sphere is isolated in space. Measurements of the electric potential V are made for different distances x from the centre of the sphere.

The variation with distance x of the potential V is shown in Fig. 5.1.

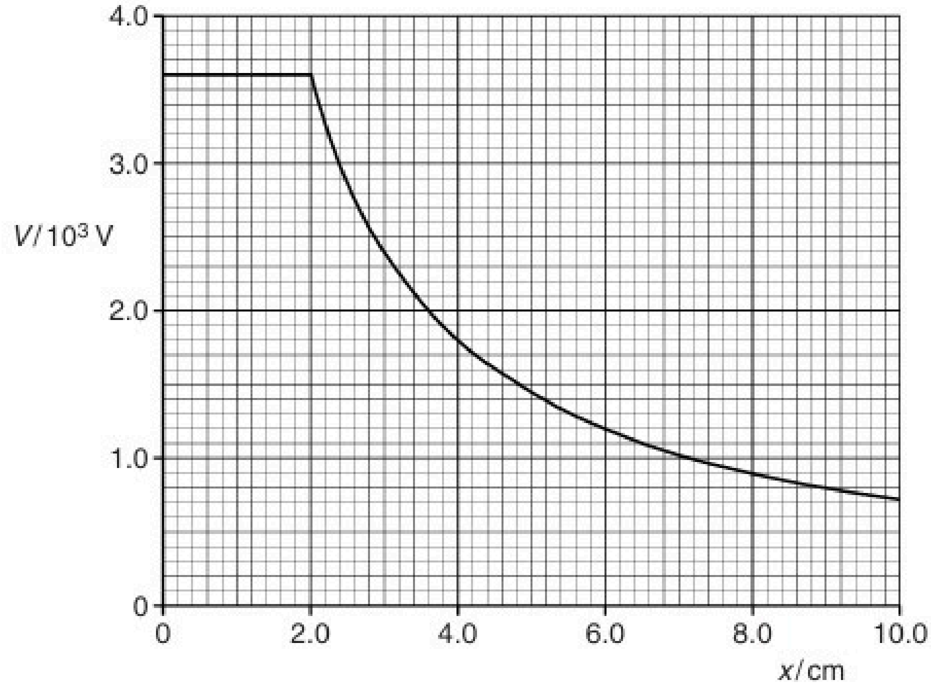


Fig. 5.1

(a) Use Fig. 5.1 to determine the electric field strength, in NC^{-1}, at a point where $x = 4.0$ cm. Explain your working.

electric field strength = ... NC^{-1} [3]

(b) The charge on the sphere is 8.0×10^{-9} C.

(i) Use Fig. 5.1 to state the electric potential at the surface of the sphere.

potential = ... V [1]

(ii) The sphere acts as a capacitor. Determine the capacitance of the sphere.

capacitance = ... F [2]

21. M/J/15/42/Q5

(a) Define *electric potential* at a point.

....

....

....[2]

(b) Two positively charged metal spheres A and B are situated in a vacuum, as shown in Fig. 5.1.

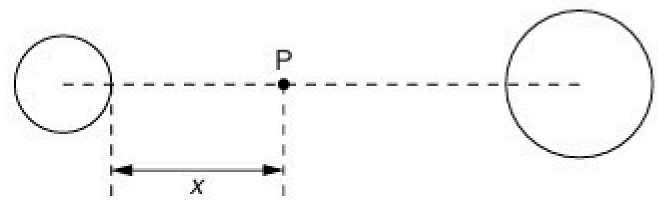
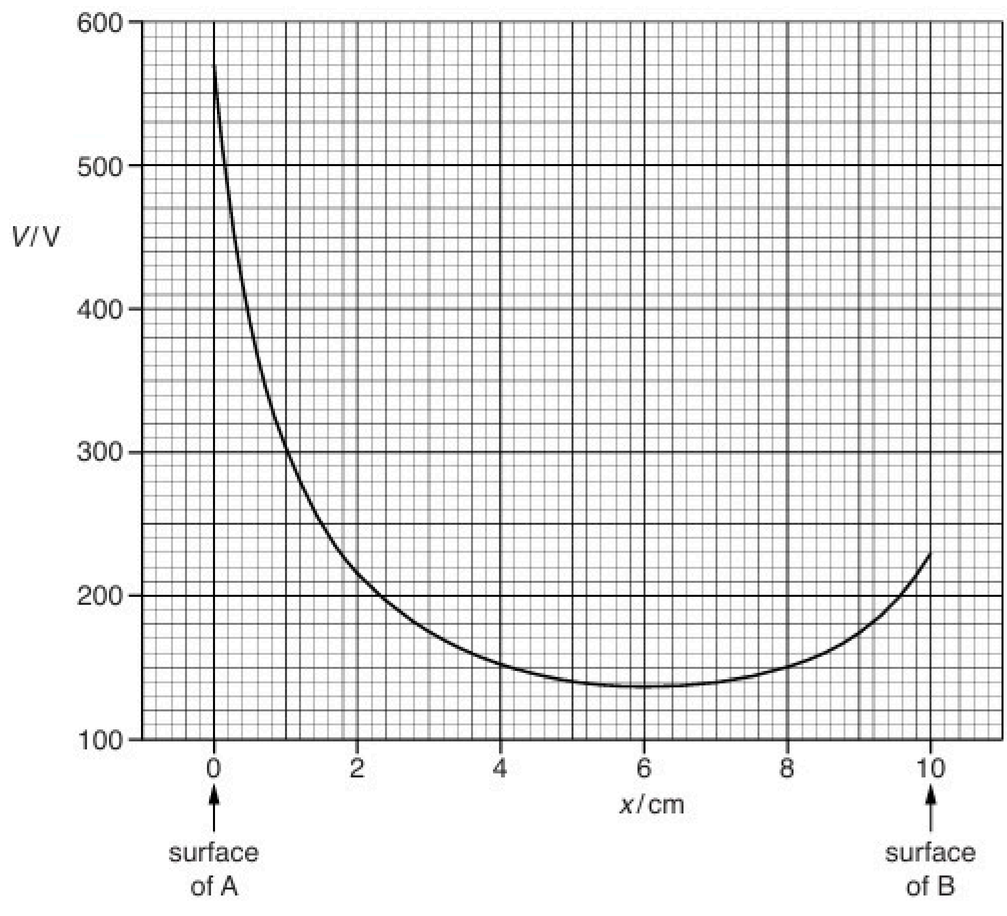


Fig. 5.1

A point P lies on the line joining the centres of the two spheres and is a distance x from the surface of sphere A.

The variation with x of the electric potential V due to the two charged spheres is shown in Fig. 5.2.



(i) State how the magnitude of the electric field strength at any point P may be determined from the graph of Fig. 5.2.

....

....[1]

(ii) Without any calculation, describe the force acting on a positively charged particle placed at point P for values of x from $x = 0$ to $x = 10\,\text{cm}$.

....

....

....

....[3]

(c) The positively charged particle in **(b)(ii)** has charge q and mass m given by the expression

$$\frac{q}{m} = 4.8 \times 10^{7}\,\text{C}\,\text{kg}^{-1}.$$

Initially, the particle is at rest on the surface of sphere A where $x = 0$. It then moves freely along the line joining the centres of the spheres until it reaches the surface of sphere B.

(i) On Fig. 5.2, mark with the letter M the point where the charged particle has its maximum speed. [1]

(ii) **1.** Use Fig. 5.2 to determine the potential difference between the spheres.

potential difference = ... V [1]

2. Use your answer in **(ii) part 1** to calculate the speed of the particle as it reaches the surface of sphere B.
Explain your working.

speed = ... $\text{m}\,\text{s}^{-1}$ [3]

22. O/N/15/41/Q5

A charged particle P is situated in a vacuum at a distance x from the centre of a charged conducting sphere of radius r, as illustrated in Fig. 5.1.

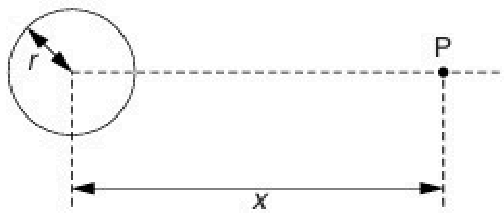


Fig. 5.1

For the particle P outside the conducting sphere, the charge on the sphere may be assumed to be a point charge at its centre.

(a) **(i)** State Coulomb's law.

....

....

....[2]

(ii) The sphere and the particle P are both charged positively.

1. State the direction of the force acting on particle P.

....[1]

2. State the position of particle P for the force to be maximum.

....[1]

3. Determine the ratio

$$\frac{\text{force on particle P at } x = r}{\text{force on particle P at } x = 4r}.$$

ratio = ... [2]

(b) When the charge on the sphere is 6.0×10^{-7} C, the electric field strength at the surface of the sphere is $1.5 \times 10^{6}\,\mathrm{V\,m^{-1}}$.

Electrical breakdown (a spark) occurs when the electric field strength at the surface of the sphere exceeds $2.0 \times 10^{6}\,\mathrm{V\,m^{-1}}$.

Determine the additional charge that may be added to the sphere before breakdown occurs.

charge = ... C [3]

23. O/N/15/43/Q5

A positively charged solid metal sphere is isolated in space. The electric field strength E is measured for different distances x from the centre of the sphere. The variation with x of the field strength E is shown in Fig. 5.1.

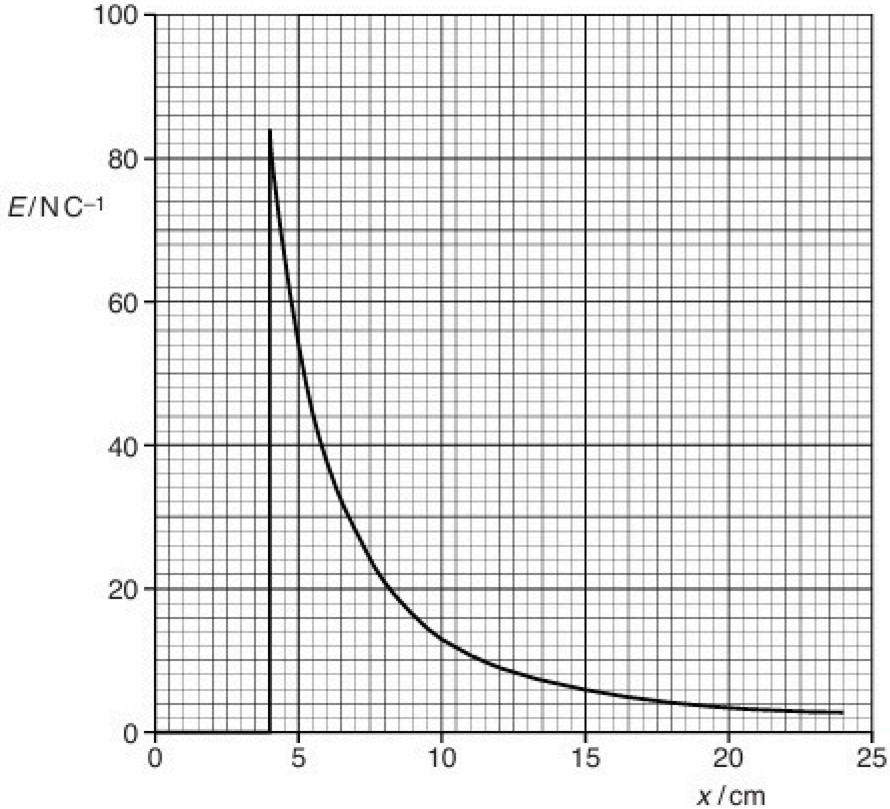


Fig. 5.1

(a) Suggest why, for values of x less than 4.0 cm, the electric field strength is zero.

....

....

.... [2]

(b) A point charge of $+8.5 \times 10^{-9}$ C moves from a point where $x = 7.0$ cm to a point where $x = 5.0$ cm.

Use Fig. 5.1 to estimate the change in electric potential energy of this point charge.

energy = ... J [3]

24. M/J/16/41/Q6

A solid metal sphere of radius R is isolated in space. The sphere is positively charged so that the electric potential at its surface is V_S. The electric field strength at the surface is E_S.

(a) On the axes of Fig. 6.1, show the variation of the electric potential with distance x from the centre of the sphere for values of x from $x = 0$ to $x = 3R$.

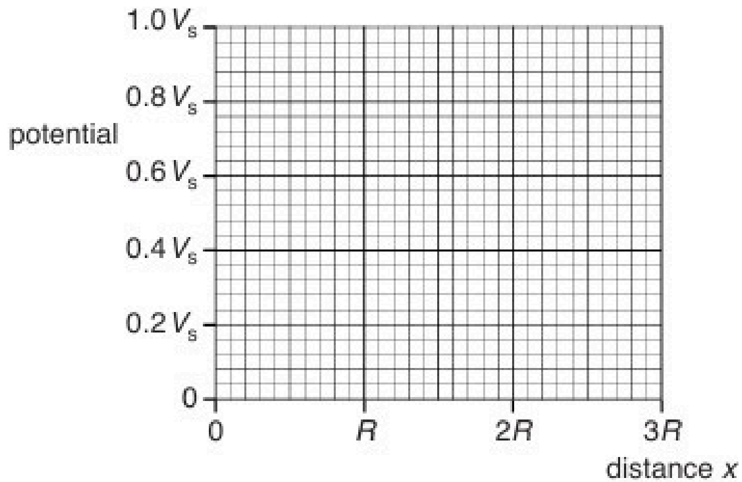


Fig. 6.1

[3]

(b) On the axes of Fig. 6.2, show the variation of the electric field strength with distance x from the centre of the sphere for values of x from $x = 0$ to $x = 3R$.

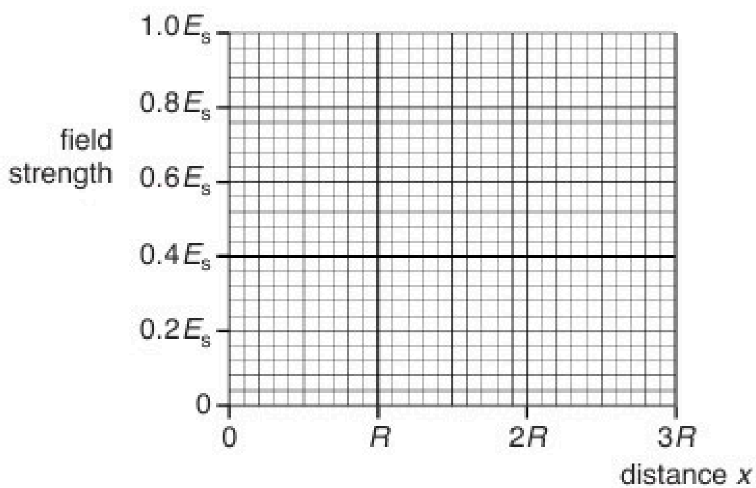


Fig. 6.2

[3]

[Total: 6]

25. M/J/16/42/Q6

(a) By reference to electric field lines, explain why, for points outside an isolated spherical conductor, the charge on the sphere may be considered to act as a point charge at its centre.

....

....

....

.... [2]

(b) Two isolated protons are separated in a vacuum by a distance *x*.

(i) Calculate the ratio

$$\frac{\text{electric force between the two protons}}{\text{gravitational force between the two protons}}.$$

ratio = ... [3]

(ii) By reference to your answer in **(i)**, suggest why gravitational forces are not considered when calculating the force between charged particles.

....

.... [1]

[Total: 6]

26. O/N/16/41/Q5

Two small solid metal spheres A and B have equal radii and are in a vacuum. Their centres are 15 cm apart.
Sphere A has charge +3.0 pC and sphere B has charge +12 pC. The arrangement is illustrated in Fig. 5.1.

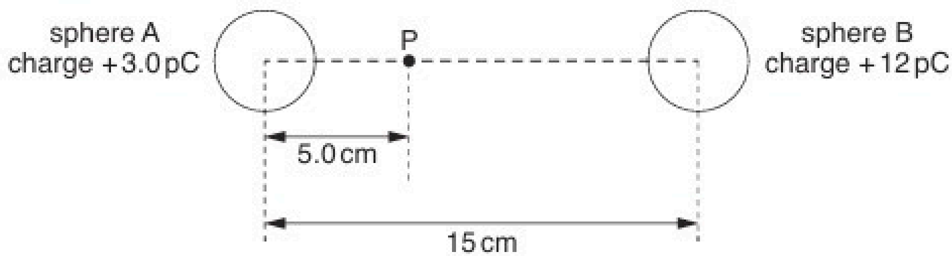


Fig. 5.1

Point P lies on the line joining the centres of the spheres and is a distance of 5.0 cm from the centre of sphere A.

(a) Suggest why the electric field strength in both spheres is zero.

....

....

.... [2]

(b) Show that the electric field strength is zero at point P. Explain your working.

[3]

(c) Calculate the electric potential at point P.

electric potential = ... V [2]

(d) A silver-107 nucleus ($^{107}_{47}Ag$) has speed v when it is a long distance from point P.

Use your answer in **(c)** to calculate the minimum value of speed v such that the nucleus can reach point P.

speed = ... $m\,s^{-1}$ [3]

[Total: 10]

27. O/N/16/42/Q6

Two solid metal spheres A and B, each of radius 1.5 cm, are situated in a vacuum. Their centres are separated by a distance of 20.0 cm, as shown in Fig. 6.1.

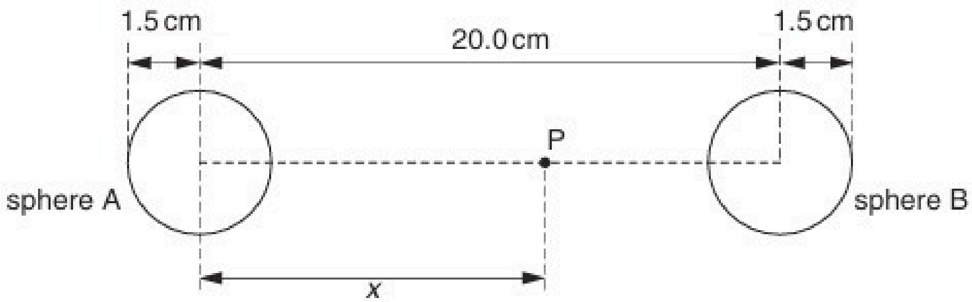


Fig. 6.1 (not to scale)

Both spheres are positively charged.

Point P lies on the line joining the centres of the two spheres, at a distance x from the centre of sphere A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 6.2.

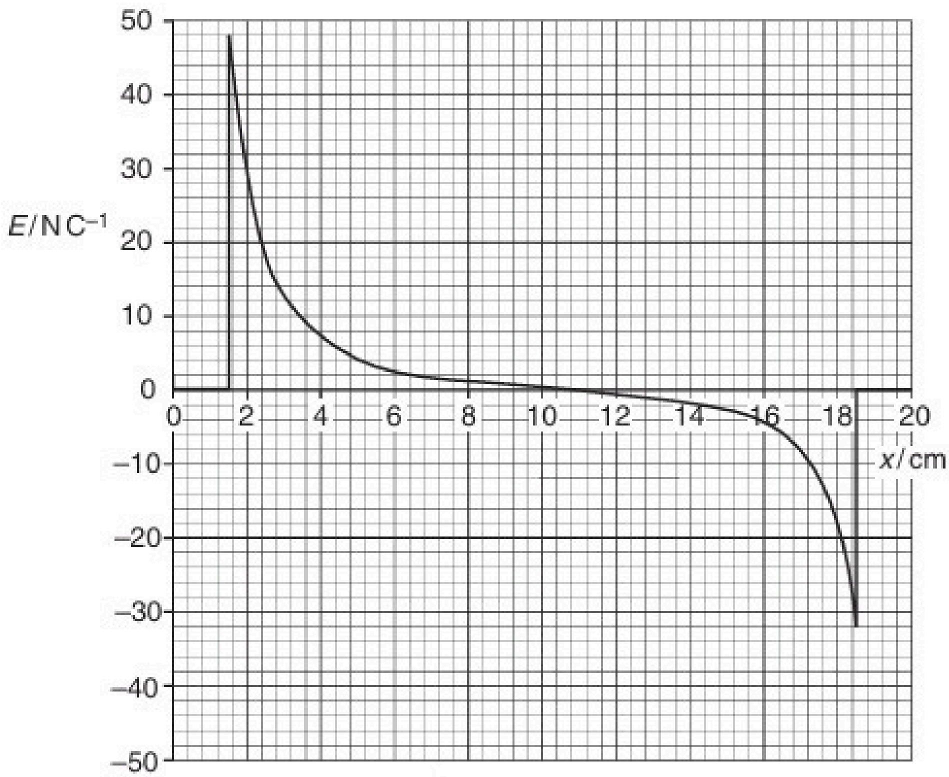


Fig. 6.2

(a) Use Fig. 6.2 to determine the ratio

$$\frac{\text{magnitude of charge on sphere A}}{\text{magnitude of charge on sphere B}}.$$

Explain your working.

ratio = ...[3]

(b) The variation with distance *x* of the electric potential *V* at point P is shown in Fig. 6.3.

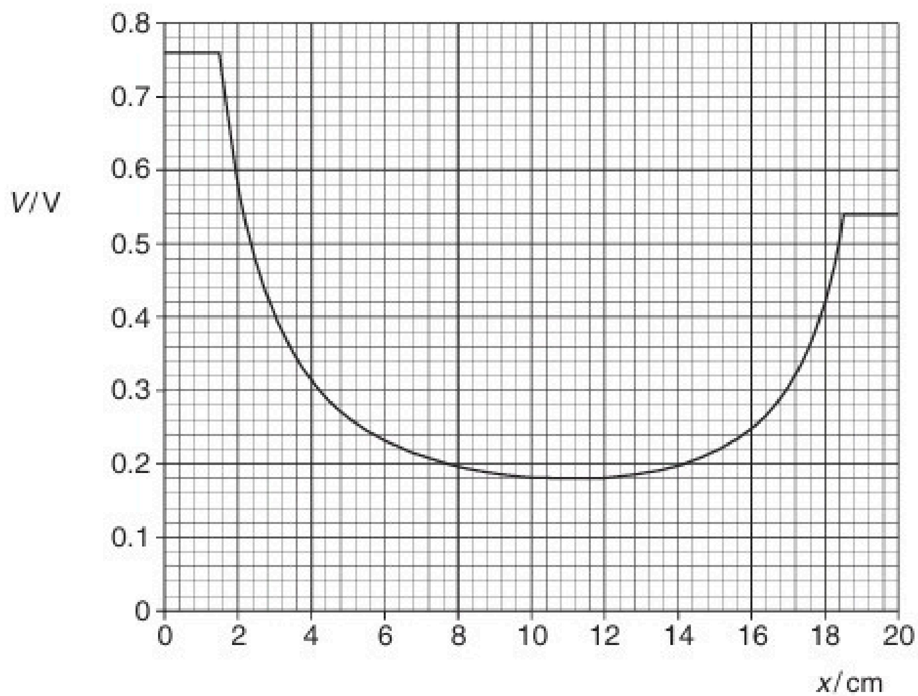


Fig. 6.3

An α-particle is initially at rest on the surface of sphere A.
The α-particle moves along the line joining the centres of the two spheres.

Determine, for the α-particle as it moves between the two spheres,

(i) its maximum speed,

maximum speed = ... $m s^{-1}$ [3]

(ii) its speed on reaching the surface of sphere B.

speed = ... $m s^{-1}$ [2]

[Total: 8]

28. M/J/17/41/Q5

An α-particle is travelling in a vacuum towards the centre of a gold nucleus, as illustrated in Fig. 5.1.

Fig. 5.1

The gold nucleus has charge 79*e*.
The gold nucleus and the α-particle may be assumed to behave as point charges.
At a large distance from the gold nucleus, the α-particle has energy 7.7×10^{-13} J.

(a) The α-particle does not collide with the gold nucleus. Show that the radius of the gold nucleus must be less than 4.7×10^{-14} m.

[3]

(b) Determine the acceleration of the α-particle for a separation of 4.7×10^{-14} m between the centres of the gold nucleus and of the α-particle.

acceleration = ... m s^{-2} [3]

(c) In an α-particle scattering experiment, the beam of α-particles is incident on a very thin gold foil.
Suggest why the gold foil must be very thin.

....

....[1]

29. M/J/17/42/Q6

(a) State Coulomb's law.

....

....

.... [2]

(b) Two charged metal spheres A and B are situated in a vacuum, as illustrated in Fig. 6.1.

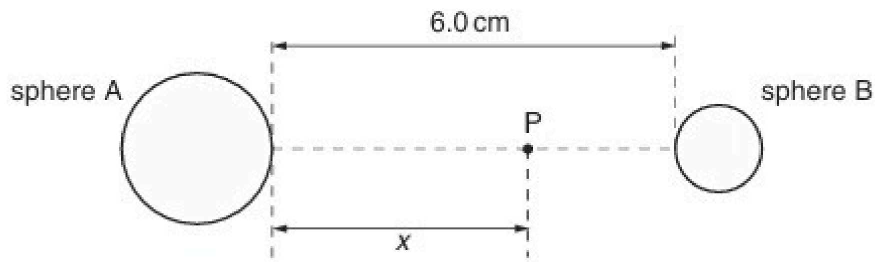


Fig. 6.1

The shortest distance between the surfaces of the spheres is 6.0 cm.

A movable point P lies along the line joining the centres of the two spheres, a distance *x* from the surface of sphere A.

The variation with distance *x* of the electric field strength *E* at point P is shown in Fig. 6.2.

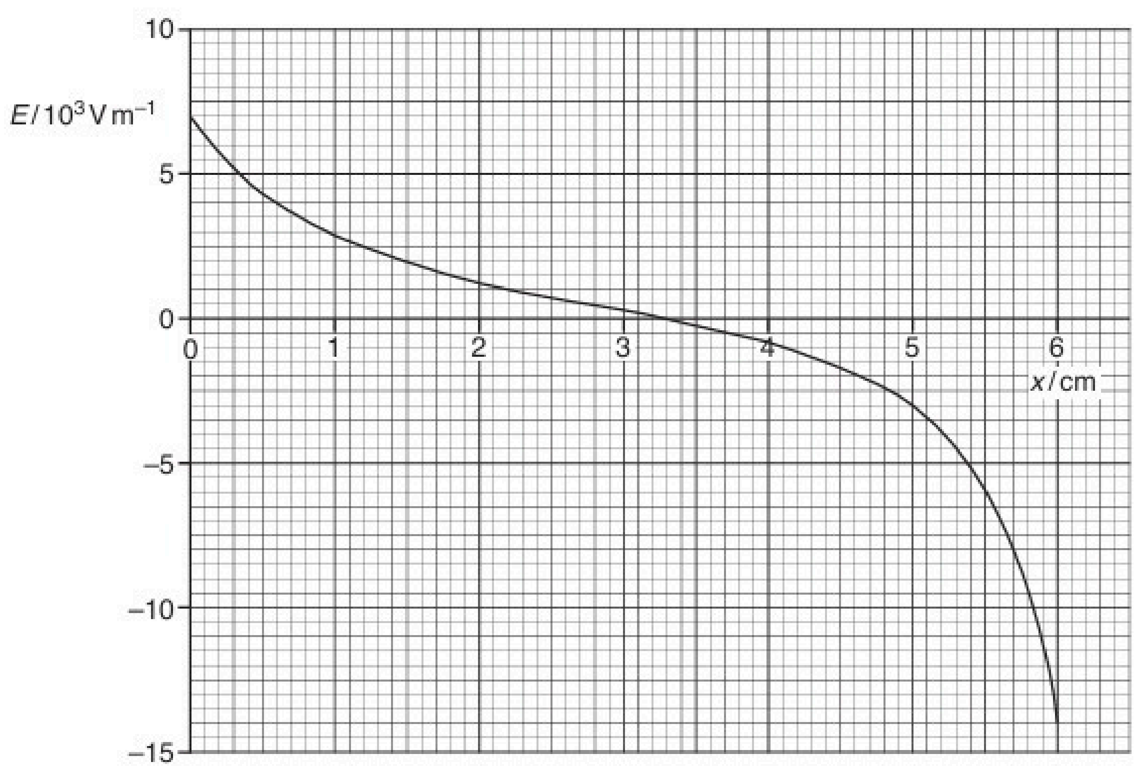


Fig. 6.2

(i) Use Fig. 6.2 to explain whether the two spheres have charges of the same, or opposite, sign.

....

....

....

....[2]

(ii) A proton is at point P where $x = 5.0\,\text{cm}$.
Use data from Fig. 6.2 to determine the acceleration of the proton.

acceleration = ... m s^{-2} [3]

(c) Use data from Fig. 6.2 to state the value of x at which the rate of change of electric potential is maximum. Give the reason for the value you have chosen.

....

....

....[2]

[Total: 9]

30. O/N/17/42/Q6

(a) For any point outside a spherical conductor, the charge on the sphere may be considered to act as a point charge at its centre. By reference to electric field lines, explain this.

....

....

....

....[2]

(b) An isolated spherical conductor has charge q, as shown in Fig. 6.1.

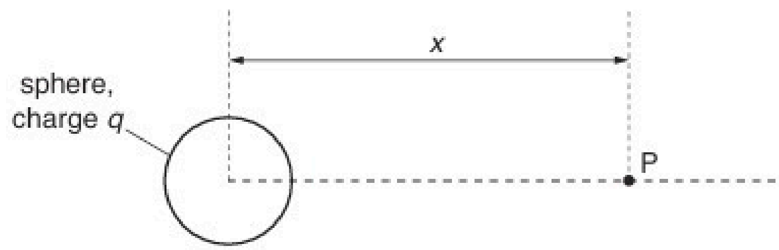


Fig. 6.1

Point P is a movable point that, at any one time, is a distance x from the centre of the sphere.

The variation with distance x of the electric potential V at point P due to the charge on the sphere is shown in Fig. 6.2.

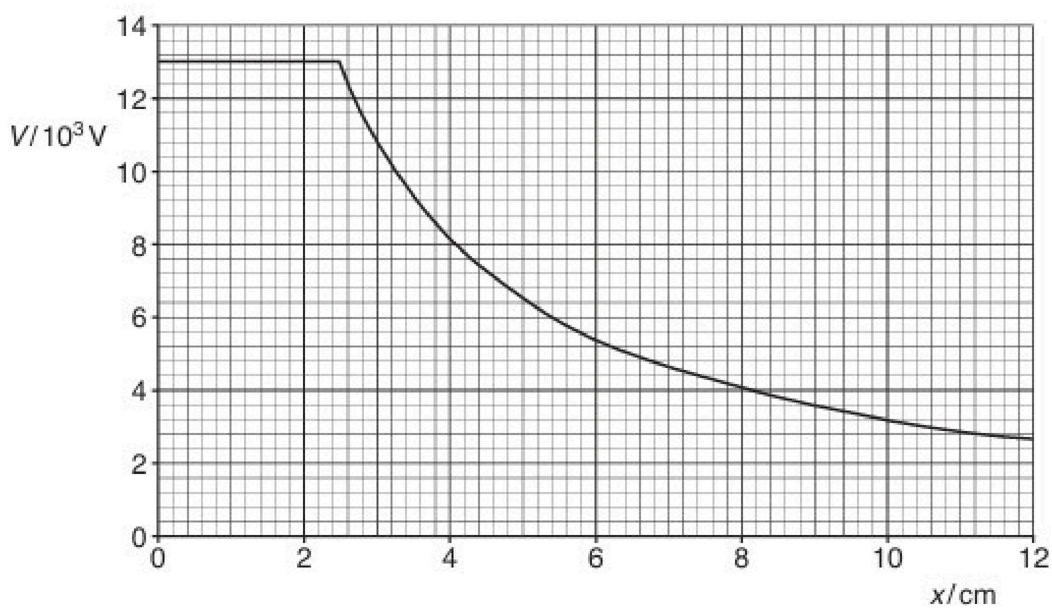


Fig. 6.2

Use Fig. 6.2 to determine

(i) the electric field strength E at point P where $x = 6.0\,\text{cm}$,

E = ... N C^{-1} [3]

(ii) the radius R of the sphere. Explain your answer.

R = ... cm [2]

[Total: 7]

31. F/M/18/42/Q7

(a) State what is meant by *electric potential* at a point.

....

....

....[2]

(b) The centres of two charged metal spheres A and B are separated by a distance of 44.0 cm, as shown in Fig. 7.1.

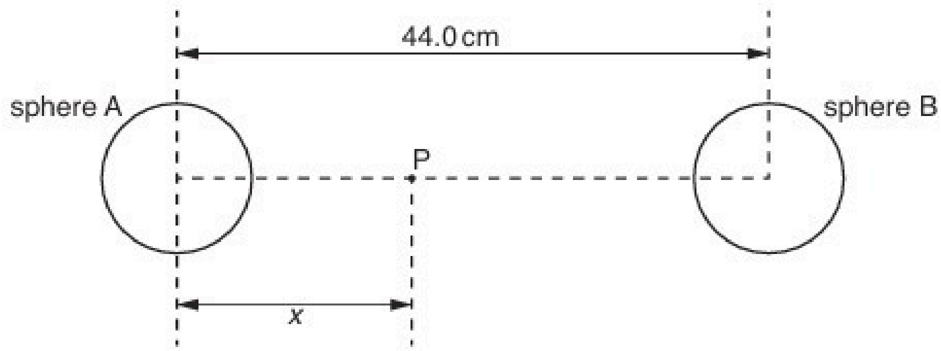
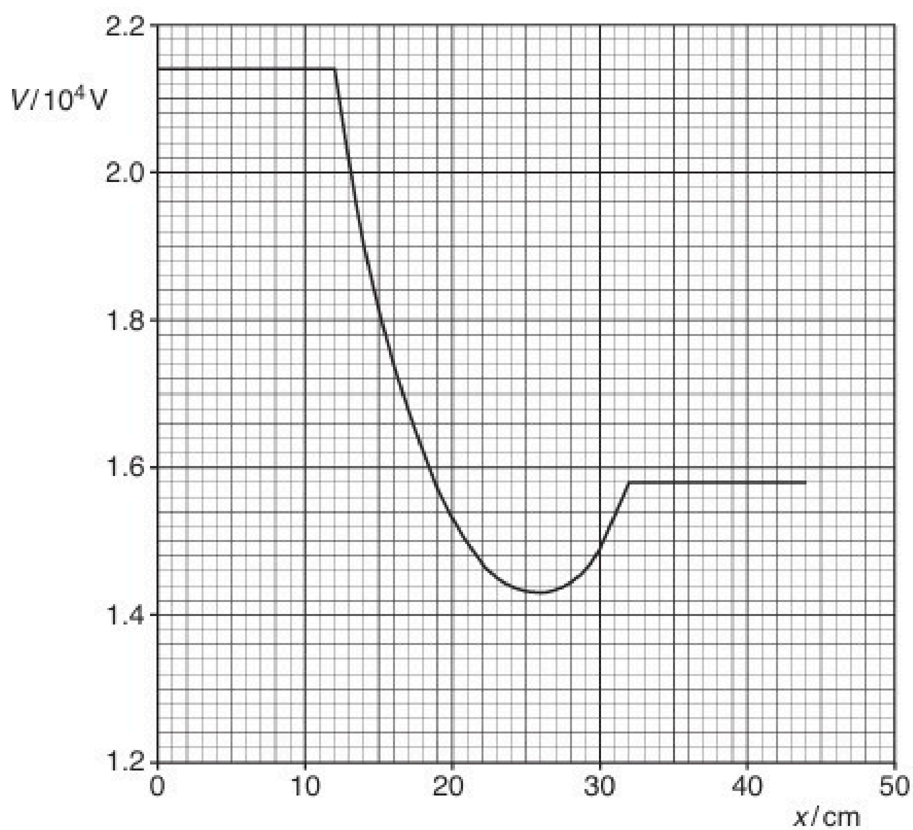


Fig. 7.1 (not to scale)

A moveable point P lies on the line joining the centres of the two spheres. Point P is a distance *x* from the centre of sphere A. The variation with distance *x* of the electric potential *V* at point P is shown in Fig. 7.2.



(i) Use Fig. 7.2 to state and explain whether the two spheres have charges of the same, or opposite, sign.

....

....

....[1]

(ii) A positively-charged particle is at rest on the surface of sphere A.

The particle moves freely from the surface of sphere A to the surface of sphere B.

1. Describe qualitatively the variation, if any, with distance *x* of the speed of the particle as it

moves from $x = 12\,\text{cm}$ to $x = 25\,\text{cm}$ ....

....

passes through $x = 26\,\text{cm}$ ....

....

moves from $x = 27\,\text{cm}$ to $x = 31\,\text{cm}$ ....

....

reaches $x = 32\,\text{cm}$ ....

....[4]

2. The particle has charge $3.2 \times 10^{-19}\,\text{C}$ and mass $6.6 \times 10^{-27}\,\text{kg}$.

Calculate the maximum speed of the particle.

speed = ... m s^{-1} [2]

[Total: 9]

32. M/J/18/41/Q6

(a) State what is meant by *electric field strength*.

....

....[1]

(b) An isolated metal sphere A of radius 26 cm is positively charged. Sphere A is shown in Fig. 6.1.

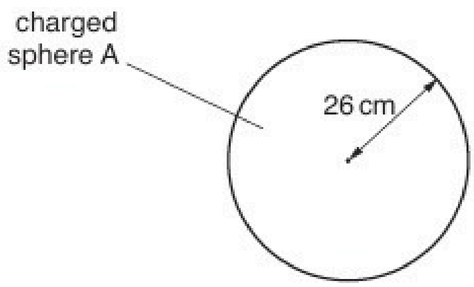


Fig. 6.1

Electrical breakdown (a spark) occurs when the electric field strength at the surface of the sphere exceeds $2.0 \times 10^4\,\mathrm{V\,m^{-1}}$.

Calculate the maximum charge Q that can be stored on the sphere.

Q = ... C [2]

(c) A second isolated metal sphere B, also with charge $+Q$, has a radius of 52 cm.

Calculate the additional charge, in terms of Q, that may be stored on this sphere before electrical breakdown occurs.

additional charge = ...[2]

33. O/N/18/41/Q6

(a) **(i)** Define *electric potential* at a point.

....

....

....[2]

(ii) State the relationship between electric potential and electric field strength at a point.

....

....

....[2]

(b) Two parallel metal plates A and B are situated a distance 1.2 cm apart in a vacuum, as shown in Fig. 6.1.

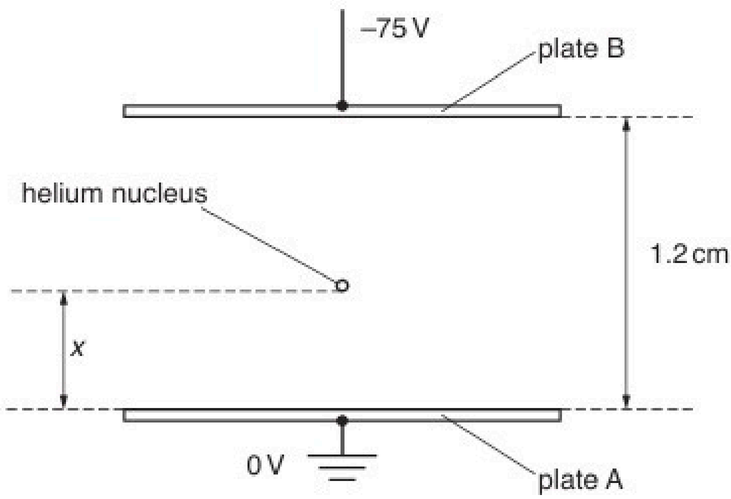


Fig. 6.1

Plate A is earthed and plate B is at a potential of –75 V.

A helium nucleus is situated between the plates, a distance *x* from plate A.

Initially, the helium nucleus is at rest on plate A where $x = 0$.

(i) The helium nucleus is free to move between the plates. By considering energy changes of the helium nucleus, explain why the speed at which it reaches plate B is independent of the separation of the plates.

....

....

....

....[2]

(ii) As the helium nucleus (^4_2He) moves from plate A towards plate B, its distance x from plate A increases.

Calculate the speed of the nucleus after it has moved a distance $x = 0.40\,cm$ from plate A.

speed = ... $m\,s^{-1}$ [3]

[Total: 9]

34. O/N/18/42/Q6

(a) State

(i) what is meant by the *electric potential* at a point,

....

....

....[2]

(ii) the relationship between electric potential at a point and electric field strength at the point.

....

....

....[2]

(b) Two similar solid metal spheres A and B, each of radius R, are situated in a vacuum such that the separation of their centres is D, as shown in Fig. 6.1.

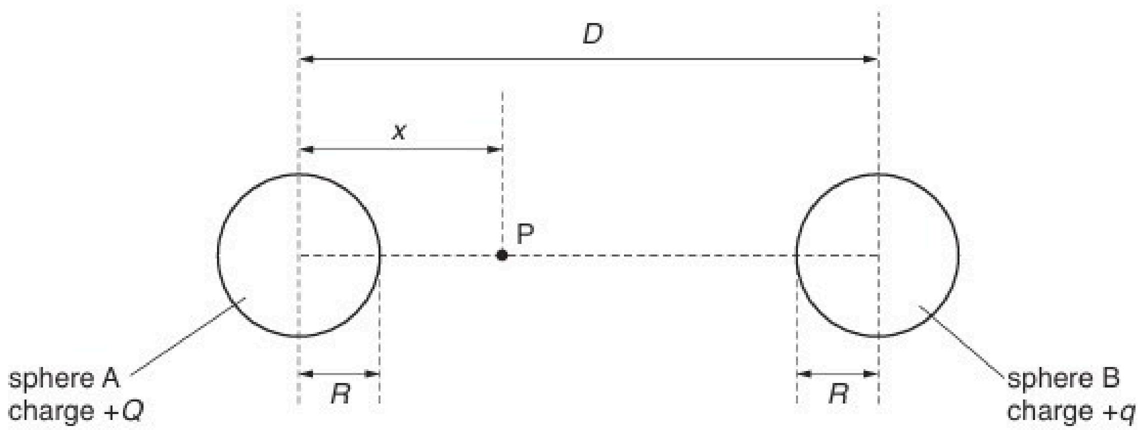


Fig. 6.1

The charge $+Q$ on sphere A is larger than the charge $+q$ on sphere B.

A movable point P is located on the line joining the centres of the two spheres.
The point P is a distance x from the centre of sphere A.

On Fig. 6.2, sketch a graph to show the variation with x of the electric potential V between the centres of the two spheres.

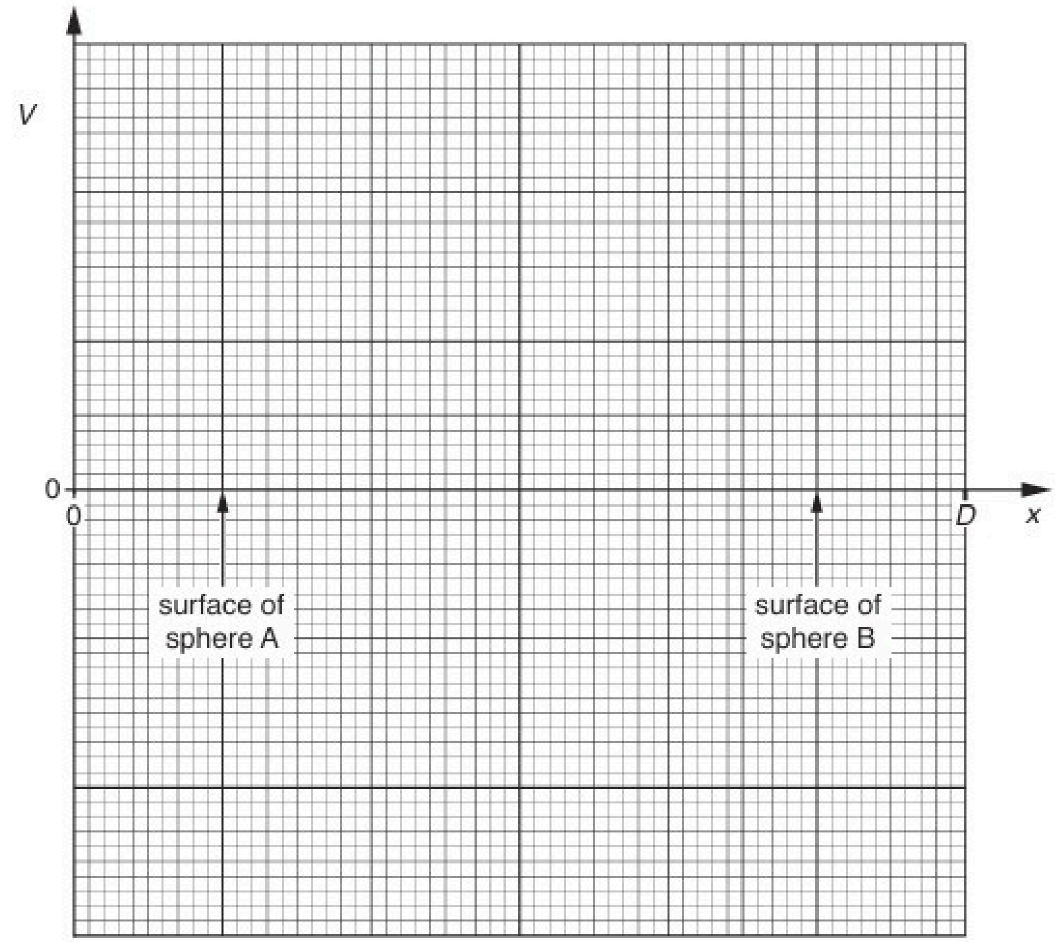


Fig. 6.2

[4]

[Total: 8]

35. F/M/19/42/Q5

(a) State what is meant by an *electric field*.

....

.... [1]

(b) An isolated solid metal sphere has radius R. The charge on the sphere is $+Q$ and the electric field strength at its surface is E.

On Fig. 5.1, draw a line to show the variation of the electric field strength with distance x from the centre of the solid sphere for values of x from $x = 0$ to $x = 3R$.

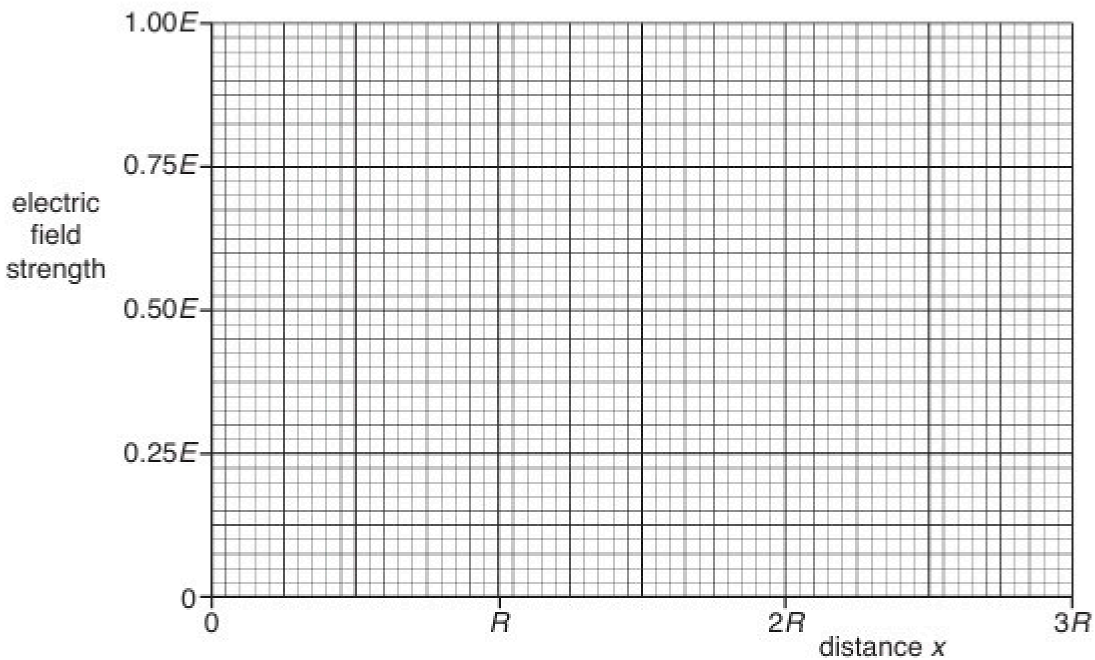


Fig. 5.1

[4]

(c) The sphere in **(b)** has radius $R = 0.26\,\text{m}$.

Electrical breakdown (a spark) occurs when the electric field strength at the surface of the sphere exceeds $2.0 \times 10^6\,\text{V m}^{-1}$.

Determine the maximum charge that can be stored on the sphere before electrical breakdown occurs.

charge = ... C [3]

36. M/J/19/41/Q5

(a) State what is meant by *electric field strength*.

....

....

.... [2]

(b) Two point charges A and B are situated a distance 15 cm apart in a vacuum, as illustrated in Fig. 5.1.

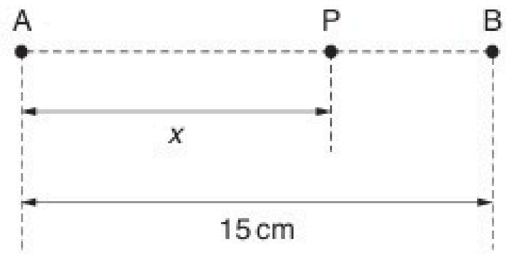


Fig. 5.1

Point P lies on the line joining the charges and is a distance x from charge A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 5.2.

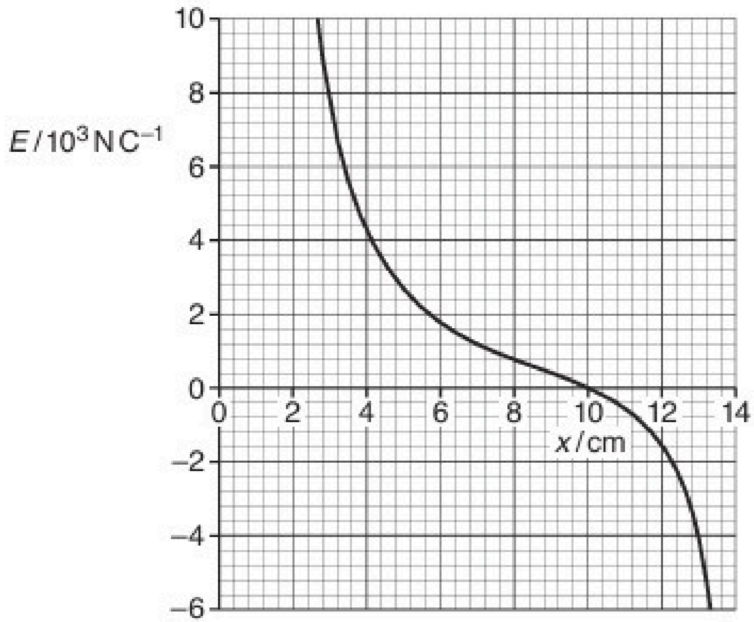


Fig. 5.2

(i) By reference to the direction of the electric field, state and explain whether the charges A and B have the same, or opposite, signs.

....

....

.... [2]

(ii) State why, although charge A is a point charge, the electric field strength between $x = 3\,\text{cm}$ and $x = 7\,\text{cm}$ does not obey an inverse-square law.

....

.... [1]

(iii) Use Fig. 5.2 to determine the ratio

$$\frac{\text{magnitude of charge A}}{\text{magnitude of charge B}}.$$

ratio = ... [3]

[Total: 8]

37. M/J/19/42/Q6

(a) State what is meant by *electric potential* at a point.

....

....

....[2]

(b) Two parallel metal plates A and B are held a distance d apart in a vacuum, as illustrated in Fig. 6.1.

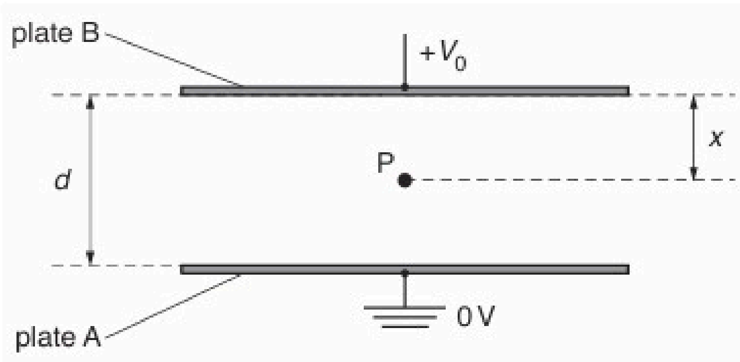


Fig. 6.1

Plate A is earthed and plate B is at a potential of $+V_0$.

Point P is situated in the centre region between the plates at a distance x from plate B. The potential at point P is V.

On Fig. 6.2, show the variation with x of the potential V for values of x from $x = 0$ to $x = d$.

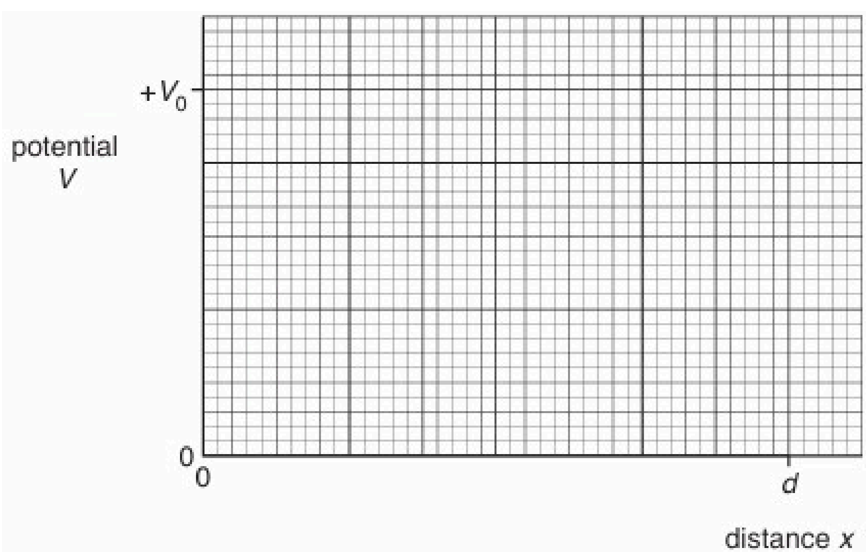


Fig. 6.2

[3]

(c) Two isolated solid metal spheres M and N, each of radius R, are situated in a vacuum. Their centres are a distance D apart, as illustrated in Fig. 6.3.

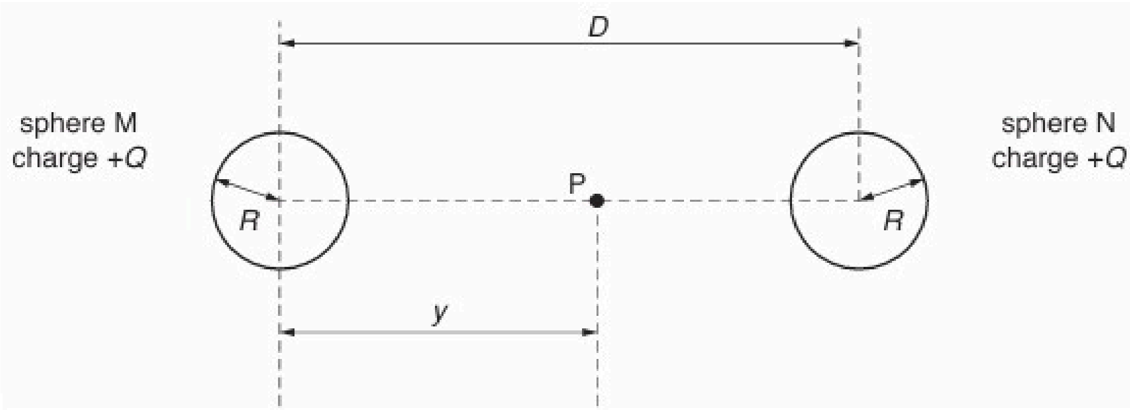


Fig. 6.3

Each sphere has charge $+Q$.

Point P lies on the line joining the centres of the two spheres, and is a distance y from the centre of sphere M.

On Fig. 6.4, show the variation with distance y of the electric potential at point P, for values of y from $y = 0$ to $y = D$.

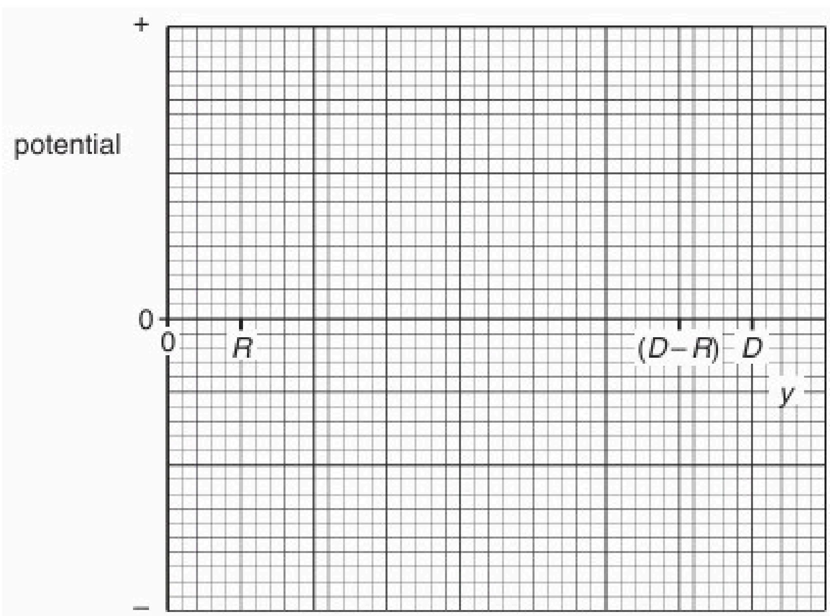


Fig. 6.4

[4]

[Total: 9]

38. O/N/19/41/Q6

(a) State an expression for the electric field strength E at a distance r from a point charge Q in a vacuum.
State the name of any other symbol used.

....

....

.... [2]

(b) Two point charges A and B are situated a distance 10.0 cm apart in a vacuum, as illustrated in Fig. 6.1.

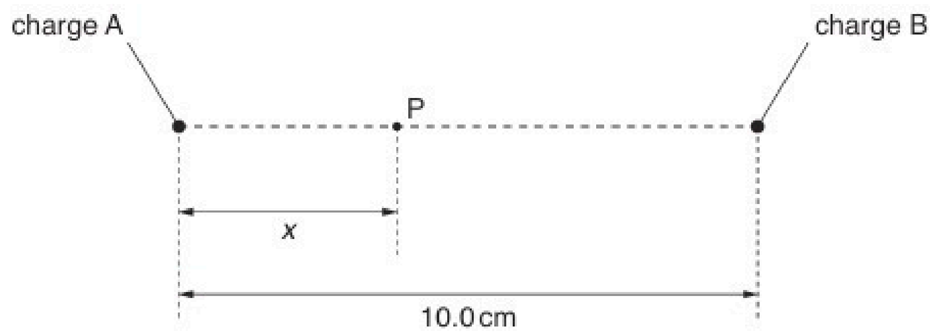


Fig. 6.1

A point P lies on the line joining the charges A and B. Point P is a distance x from A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 6.2.

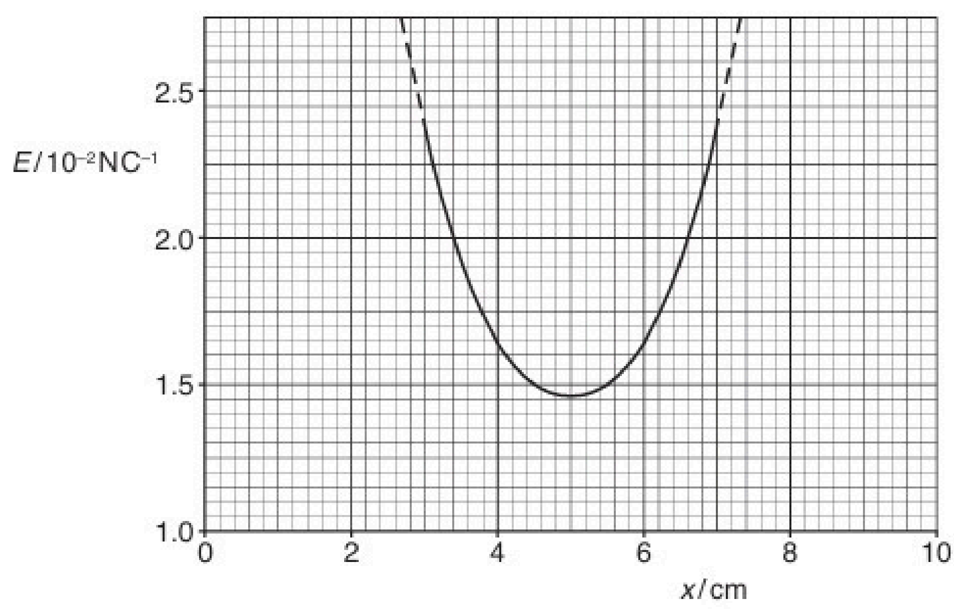


Fig. 6.2

State and explain whether the charges A and B:

(i) have the same, or opposite, signs

....

....

.... [2]

(ii) have the same, or different, magnitudes.

....

....

.... [2]

(c) An electron is situated at point P.

Without calculation, state and explain the variation in the magnitude of the acceleration of the electron as it moves from the position where $x = 3\,\text{cm}$ to the position where $x = 7\,\text{cm}$.

....

....

....

....

....

.... [4]

[Total: 10]

39. O/N/19/42/Q9

(a) Define what is meant by *electric potential* at a point.

....

....

.... [2]

(b) In an α-particle scattering experiment, α-particles are directed towards a thin film of gold, as illustrated in Fig. 9.1.

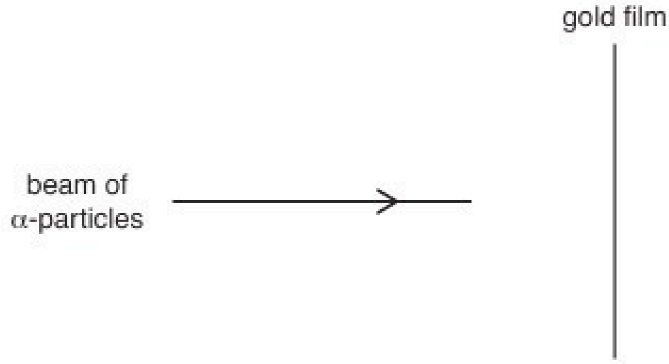


Fig. 9.1

The apparatus is in a vacuum.

The gold-197 ($^{197}_{79}Au$) nuclei in the film may be considered to be fixed point charges.

The α-particles emitted from the source each have an energy of 4.8 MeV.

Calculate:

(i) the initial kinetic energy E_K, in J, of an α-particle emitted from the source

E_K = ... J [1]

(ii) the distance d of closest approach of an α-particle to a gold nucleus.

d = ... m [4]

(c) Use your answer in **(b)(ii)** to comment on the possible diameter of a gold nucleus.

....

.... [1]

[Total: 8]

40. F/M/20/42/Q6

Two positively charged identical metal spheres A and B have their centres separated by a distance of 24 cm, as shown in Fig. 6.1.

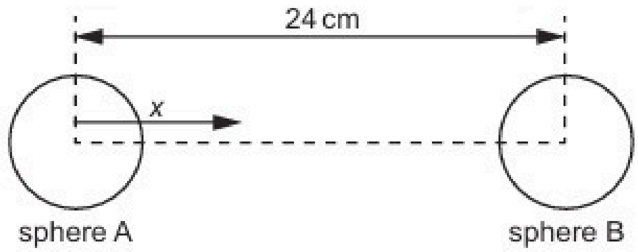


Fig. 6.1 (not to scale)

The variation with distance x from the centre of A of the electric field strength E due to the two spheres, along the line joining their centres, is represented in Fig. 6.2.

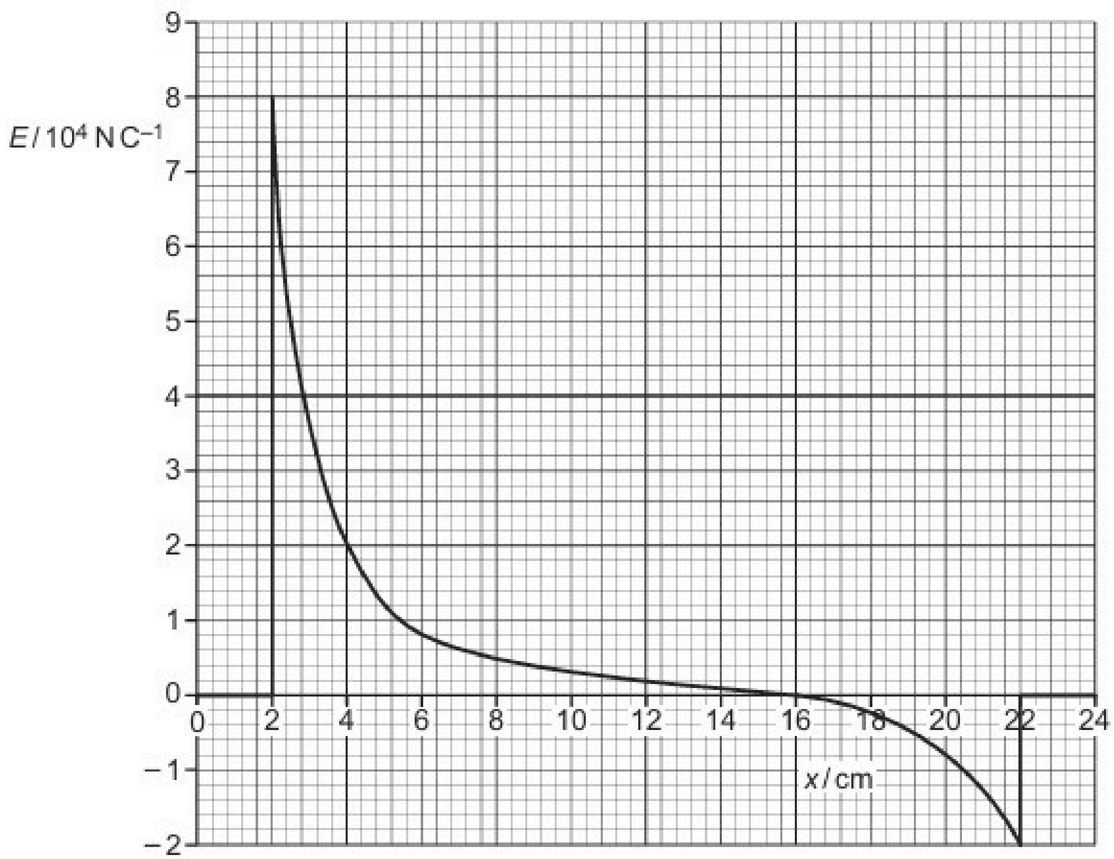


Fig. 6.2

(a) State the radius of the two spheres.

radius = ... cm [1]

(b) The charge on sphere A is 3.6×10^{-9} C. Determine the charge Q_B on sphere B.

Assume that spheres A and B can be treated as point charges at their centres.

Explain your working.

Q_B = ... C [3]

(c) **(i)** Sphere B is removed.

Use information from **(b)** to determine the electric potential on the surface of sphere A.

electric potential = ... V [2]

(ii) Calculate the capacitance of sphere A.

capacitance = ... F [2]

[Total: 8]

41. M/J/20/41/Q5

(a) State **one** similarity and **one** difference between the fields of force produced by an isolated point charge and by an isolated point mass.

similarity: ....

....

difference: ....

....

[2]

(b) An isolated solid metal sphere A of radius R has charge $+Q$, as illustrated in Fig. 5.1.

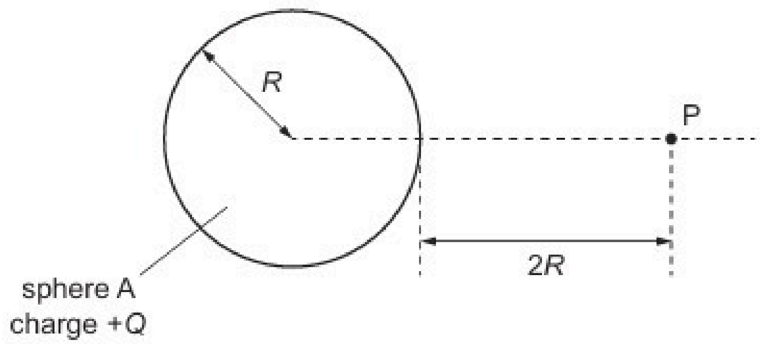


Fig. 5.1

A point P is distance $2R$ from the surface of the sphere.

Determine an expression that includes the terms R and Q for the electric field strength E at point P.

E = ... [2]

(c) A second identical solid metal sphere B is now placed near sphere A. The centres of the spheres are separated by a distance $6R$, as shown in Fig. 5.2.

Fig. 5.2

Point P lies midway between spheres A and B.

Sphere B has charge $-Q$.

Explain why:

(i) the magnitude of the electric field strength at P is given by the sum of the magnitudes of the field strengths due to each sphere

....

.... [1]

(ii) the electric field strength at point P due to the charged metal spheres is not, in practice, equal to $2E$, where E is the electric field strength determined in **(b)**.

....

....

.... [2]

[Total: 7]

42. M/J/20/42/Q7

A metal sphere of radius R is isolated in space.

Point P is a distance x from the centre of the sphere, as illustrated in Fig. 7.1.

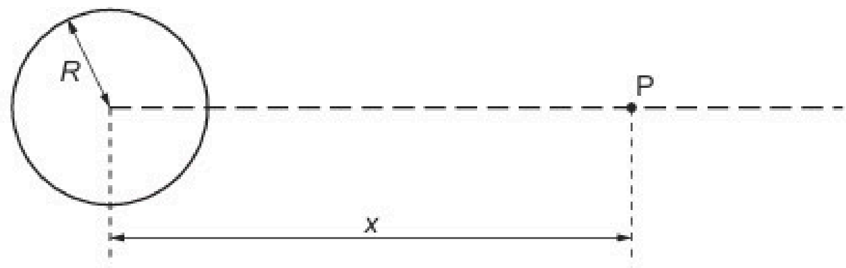


Fig. 7.1

The variation with distance x of the electric field strength E due to the charge on the sphere is shown in Fig. 7.2.

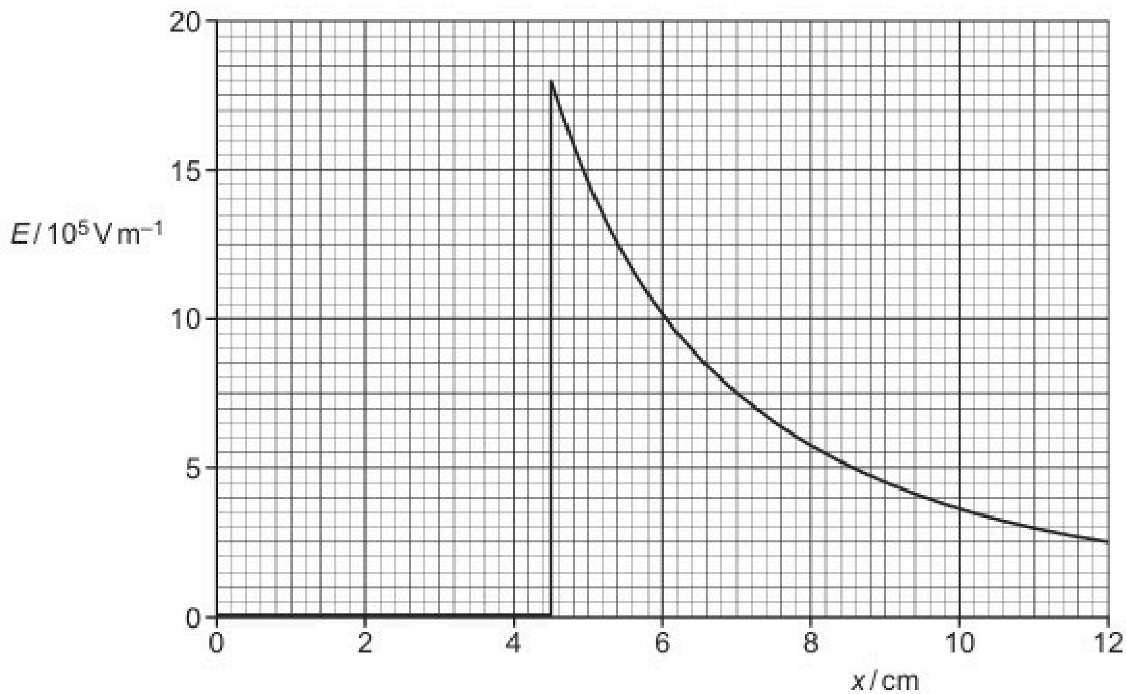


Fig. 7.2

(a) State what is meant by *electric field strength*.

....

....

.... [2]

(b) **(i)** Use Fig. 7.2 to determine the radius R of the sphere. Explain your working.

R = ... cm [2]

(ii) Use Fig. 7.2 to determine the charge Q on the sphere.

Q = ... C [3]

(c) An α-particle is situated a distance 8.0 cm from the centre of the sphere.

Calculate the acceleration of the α-particle.

acceleration = ... $m\,s^{-2}$ [3]

[Total: 10]

43. O/N/20/41/Q5

(a) Define *electric potential* at a point.

....

....

.... [2]

(b) Two point charges A and B are separated by a distance of 12.0 cm in a vacuum, as illustrated in Fig. 5.1.

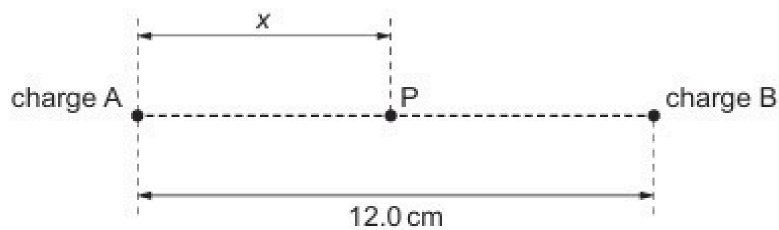
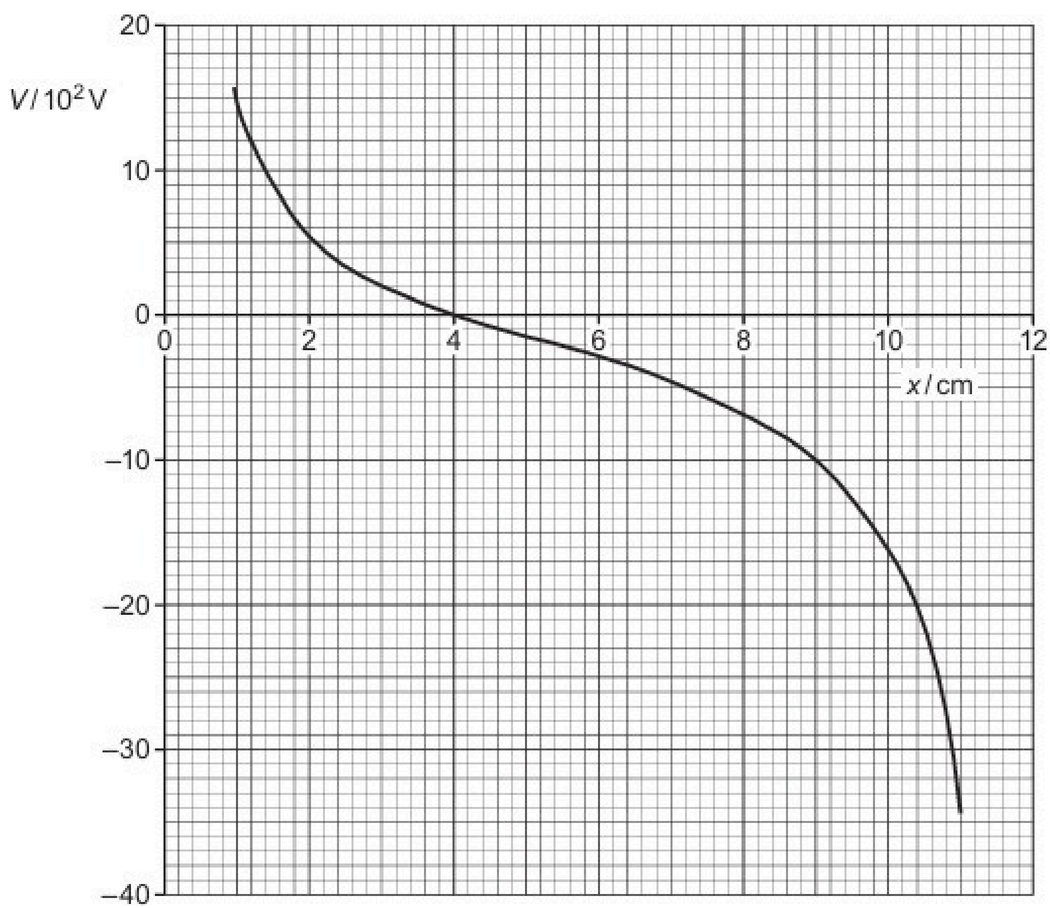


Fig. 5.1

The charge of A is $+2.0 \times 10^{-9}$ C.

A point P lies on the line joining charges A and B. Its distance from charge A is x.

The variation with distance x of the electric potential V at point P is shown in Fig. 5.2.



Use Fig. 5.2 to determine:

(i) the charge of B

charge = ... C [3]

(ii) the change in electric potential when point P moves from the position where x = 9.0 cm to the position where x = 3.0 cm.

change = ... V [1]

(c) An α-particle moves along the line joining point charges A and B in Fig. 5.1.

The α-particle moves from the position where x = 9.0 cm and just reaches the position where x = 3.0 cm.

Use your answer in **(b)(ii)** to calculate the speed v of the α-particle at the position where x = 9.0 cm.

v = ... $m\,s^{-1}$ [3]

[Total: 9]

44. O/N/20/42/Q5

(a) **(i)** State what is meant by a *field of force*.

....

....

.... [2]

(ii) State **one** similarity and **one** difference between the electric field due to a point charge and the gravitational field due to a point mass.

similarity: ....

....

....

difference: ....

....

....
[2]

(b) An isolated solid metal sphere of radius 0.15 m is situated in a vacuum, as illustrated in Fig. 5.1.

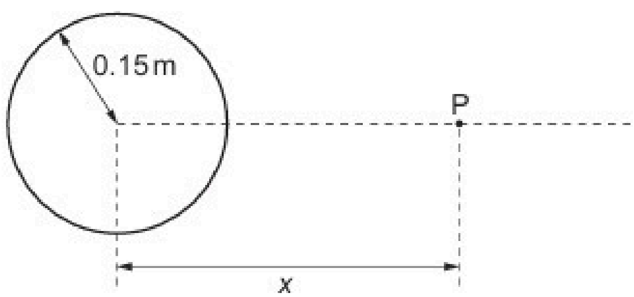


Fig. 5.1

The electric field strength at the surface of the sphere is $84\,\mathrm{V\,m^{-1}}$.

Determine:

(i) the charge Q on the sphere

Q = ... C [2]

(ii) the electric field strength at point P, a distance $x = 0.45\,\text{m}$ from the centre of the sphere.

electric field strength = ... V m^{-1} [2]

(c) Use information from **(b)** to show, on the axes of Fig. 5.2, the variation of the electric field strength E with distance x from the centre of the sphere for values of x from $x = 0$ to $x = 0.45\,\text{m}$.

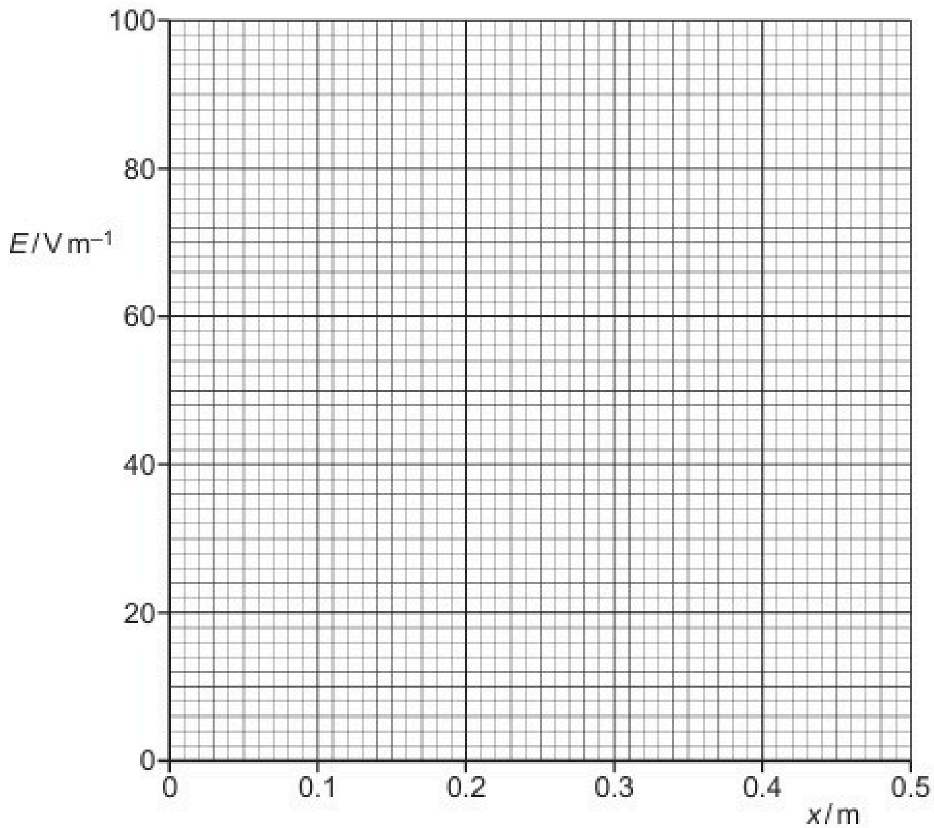


Fig. 5.2

[3]

[Total: 11]

45. F/M/21/42/Q6

(a) State a similarity between the gravitational field lines around a point mass and the electric field lines around a point charge.

....

.... [1]

(b) The variation with radius r of the electric field strength E due to an isolated charged sphere in a vacuum is shown in Fig. 6.1.

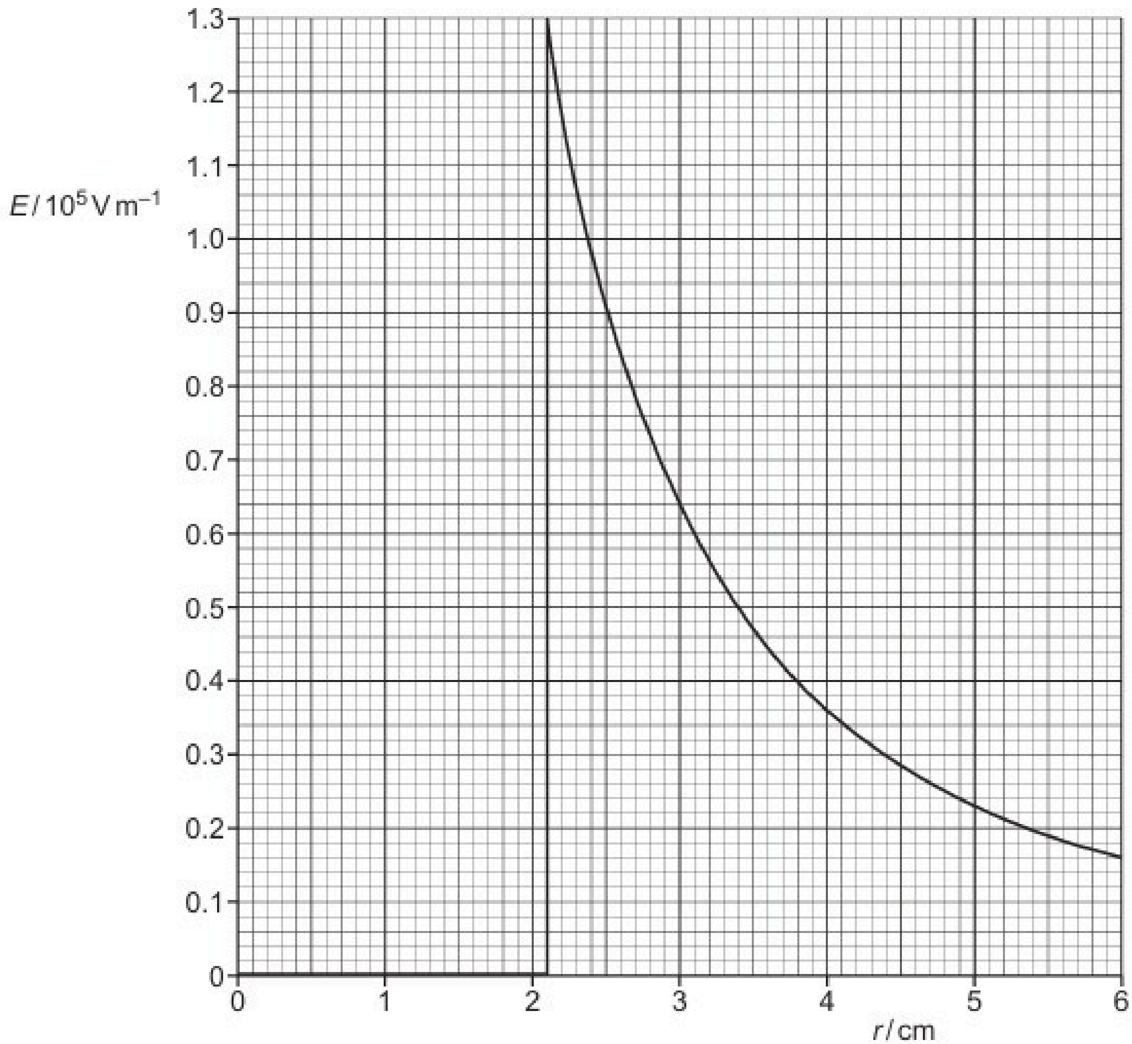


Fig. 6.1

Use data from Fig. 6.1 to:

(i) state the radius of the sphere

radius = ... cm [1]

(ii) calculate the charge on the sphere.

charge = ... C [2]

(c) Using the formula for the electric potential due to an isolated point charge, determine the capacitance of the sphere in **(b)**.

capacitance = ... F [3]

[Total: 7]

46. F/M/22/42/Q4

(a) State what is represented by an electric field line.

....

.... [2]

(b) Two point charges P and Q are placed 0.120 m apart as shown in Fig. 4.1.

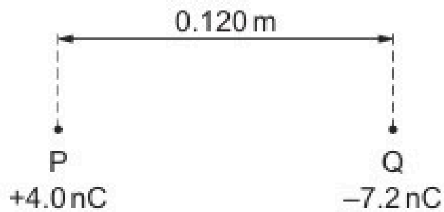


Fig. 4.1

(i) The charge of P is +4.0 nC and the charge of Q is –7.2 nC.

Determine the distance from P of the point on the line joining the two charges where the electric potential is zero.

distance = ... m [2]

(ii) State and explain, without calculation, whether the electric field strength is zero at the same point at which the electric potential is zero.

....

....

.... [1]

(iii) An electron is positioned at point X, equidistant from both P and Q, as shown in Fig. 4.2.

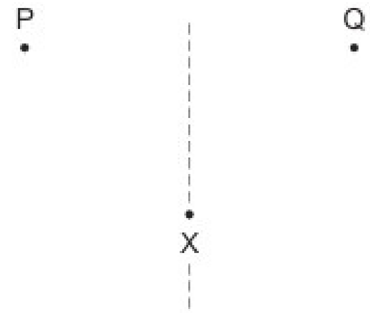


Fig. 4.2

On Fig. 4.2, draw an arrow to represent the direction of the resultant force acting on the electron. [1]

[Total: 6]

47. M/J/22/41/Q2

A sphere of mass 1.6×10^{-10} kg has a charge of +0.27 nC. The sphere is in a uniform electric field that acts vertically upwards, as shown in the side view in Fig. 2.1.

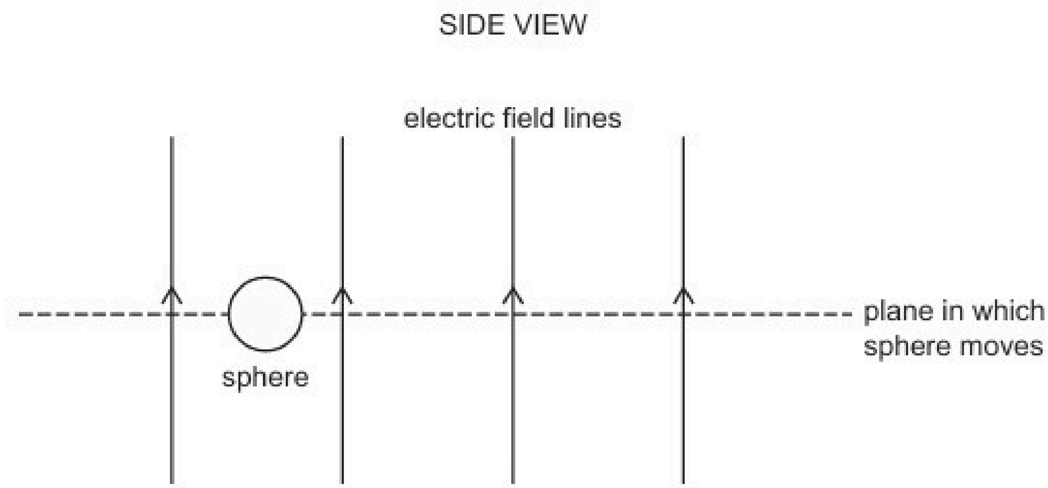


Fig. 2.1

The force exerted on the sphere by the electric field causes the sphere to remain at a fixed vertical height in a horizontal plane.

There is a uniform magnetic field in the region of the electric field. The sphere moves at a speed of $0.78\,\mathrm{m\,s^{-1}}$ in the horizontal plane. The magnetic field causes the sphere to move in a circular path of radius 3.4 m, as shown in the view from above in Fig. 2.2.

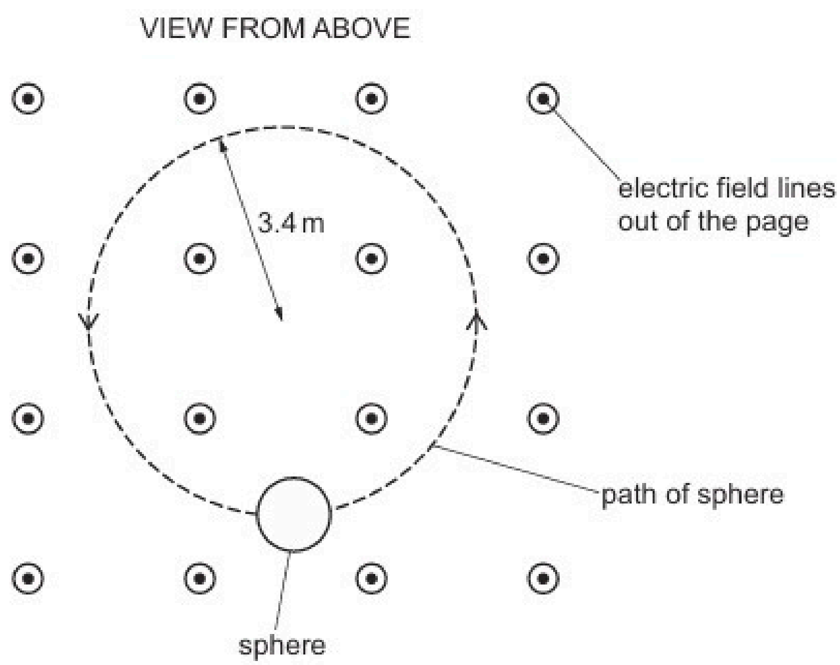


Fig. 2.2

(a) **(i)** Determine the direction of the uniform magnetic field.

.... [1]

(ii) Explain why the motion of the sphere in the horizontal plane is circular.

....

....

.... [2]

(b) Calculate the strength of the uniform electric field.

electric field strength = ... NC^{-1} [2]

(c) By considering the magnetic force on the sphere, show that the flux density of the uniform magnetic field is 0.14 T.

[3]

[Total: 8]

48. M/J/22/42/Q2

(a) State Coulomb's law.

....

....

.... [2]

(b) Positronium is a system in which an electron and a positron orbit, with the same period, around their common centre of mass, as shown in Fig. 2.1.

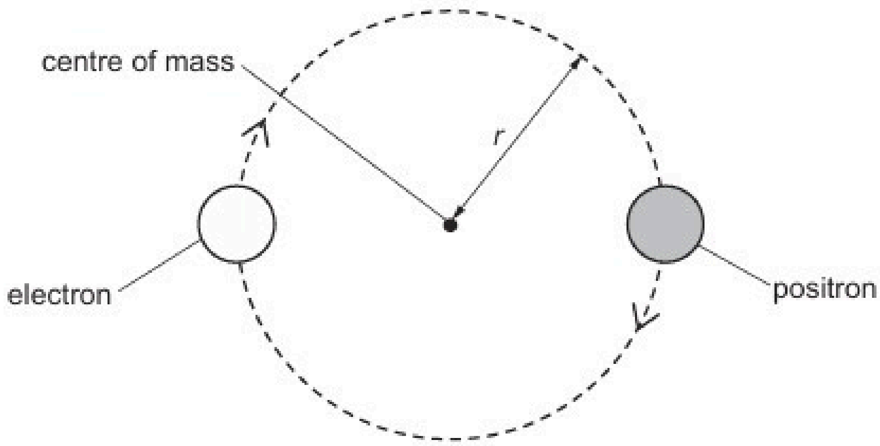


Fig. 2.1 (not to scale)

The radius r of the orbit of both particles is 1.59×10^{-10} m.

(i) Explain how the electric force between the electron and the positron causes the path of the moving particles to be circular.

....

....

.... [2]

(ii) Show that the magnitude of the electric force between the electron and the positron is 2.28×10^{-9} N.

[2]

(iii) Use the information in **(b)(ii)** to determine the period of the circular orbit of the two particles.

period = ... s [3]

(c) Positronium is highly unstable, and after a very short period of time it becomes gamma radiation.

(i) Describe how gamma radiation is formed from the two particles in positronium.

....

....

....

.... [3]

(ii) State **one** medical application of the process described in **(c)(i)**.

.... [1]

[Total: 13]

49. O/N/22/41/Q4

(a) State what is indicated by the direction of an electric field line.

....

.... [2]

(b) Fig. 4.1 shows a pair of parallel metal plates with a potential difference (p.d.) of 2400 V between them.

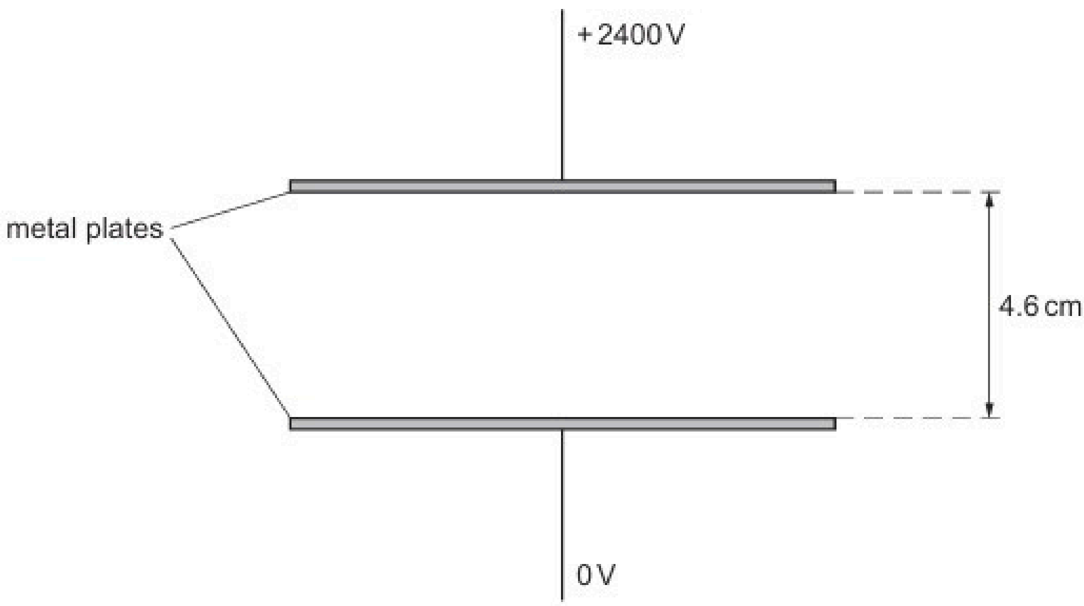


Fig. 4.1

The plates are separated by a distance of 4.6 cm. The plates are in a vacuum.

(i) On Fig. 4.1, draw five lines to represent the electric field in the region between the plates. [3]

(ii) Calculate the strength of the electric field between the plates.

electric field strength = ... NC^{-1} [2]

(c) A moving proton enters the region between the plates from the left, as shown in Fig. 4.2.

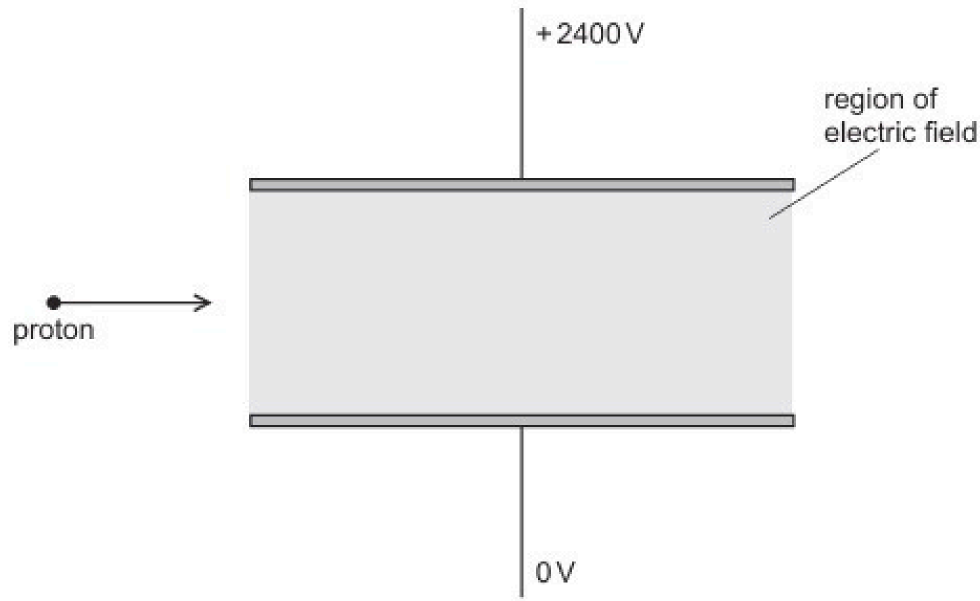


Fig. 4.2

(i) The proton is deflected by the electric field.

On Fig. 4.2, draw a line to show the path of the proton as it moves through and out of the region of the electric field. [2]

(ii) A helium nucleus ($^{4}_{2}He$) now enters the region of the electric field along the same initial path as the proton and travelling at the same initial speed.

State and explain how the final speed of the helium nucleus compares with the final speed of the proton after leaving the region of the electric field.

....

....

....

....

.... [3]

[Total: 12]

50. O/N/22/42/Q5

(a) Define electric potential at a point.

....

....

.... [2]

(b) An isolated conducting sphere is charged. Fig. 5.1 shows the variation of the potential V due to the sphere with displacement x from its centre.

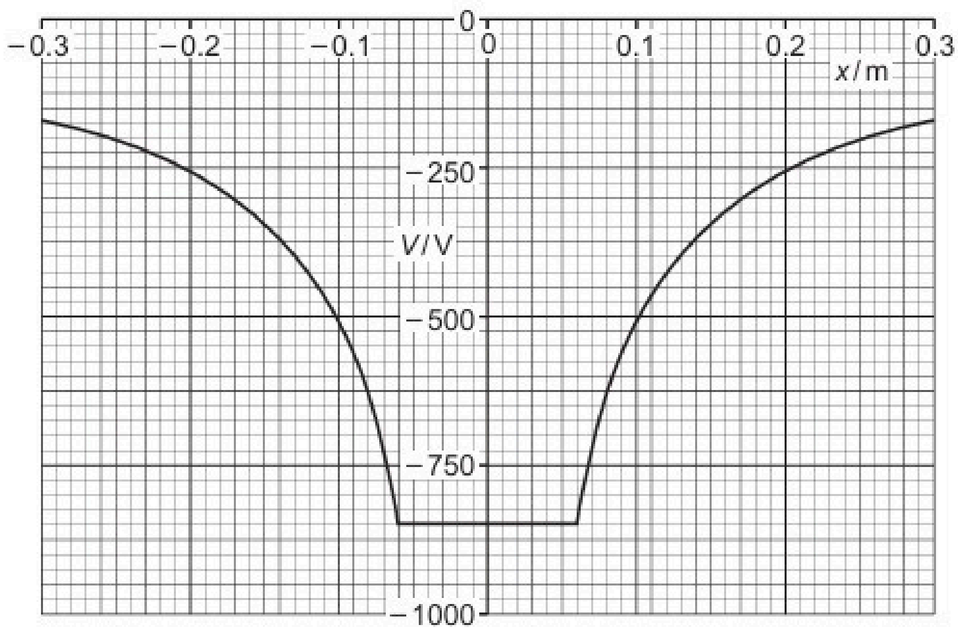


Fig. 5.1

Use Fig. 5.1 to determine:

(i) the radius of the sphere

radius = ... m [1]

(ii) the charge on the sphere.

charge = ... C [2]

(c) Two spheres are identical to the sphere in **(b)**. Each sphere has the same charge as the sphere in **(b)**.

The spheres are held in a vacuum so that their centres are separated by a distance of 0.46 m. Assume that the charge on each sphere is a point charge at the centre of the sphere.

(i) Calculate the electric potential energy E_P of the two spheres.

E_P = ... J [2]

(ii) The two spheres are now released simultaneously so that they are free to move.

Describe and explain the subsequent motion of the spheres.

....

....

....

.... [3]

[Total: 10]

51. F/M/23/42/Q4

(a) State Coulomb's law.

....

....

....

.... [2]

(b) A charged sphere X is supported on an insulating stand. A second charged sphere Y is suspended by an insulating thread so that sphere Y is in equilibrium at the position shown in Fig. 4.1.

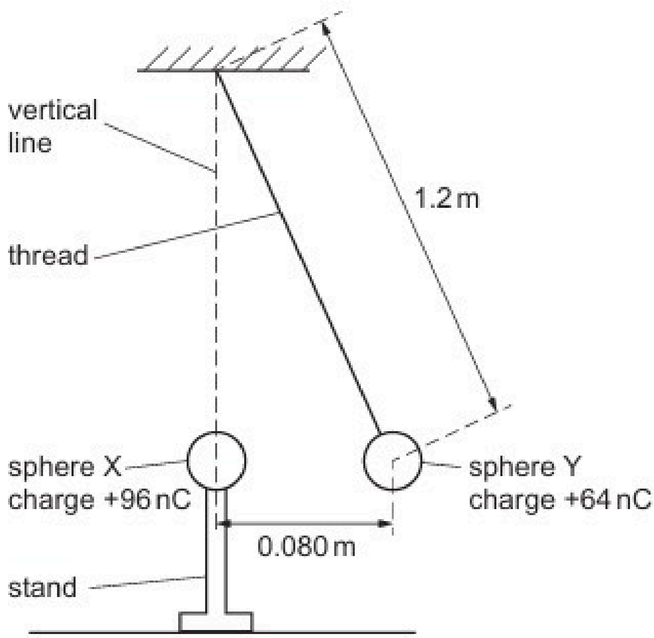


Fig. 4.1

The charge on sphere X is +96 nC and the charge on sphere Y is +64 nC.
Assume that the spheres behave as point charges.

The length of the thread is 1.2 m and the centres of sphere X and sphere Y are separated horizontally by a distance of 0.080 m.

(i) On Fig. 4.2, draw and label all the forces acting on sphere Y.

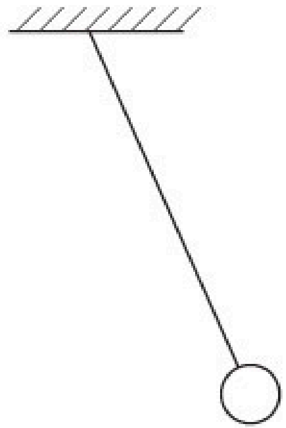


Fig. 4.2

[1]

(ii) Determine the mass of sphere Y.

mass = ... kg [4]

(iii) Calculate the total electric potential energy stored between X and Y.

energy = ... J [1]

52. M/J/23/41/Q1

(a) **(i)** Define gravitational field.

....

.... [1]

(ii) Define electric field.

....

.... [1]

(iii) State **one** similarity and **one** difference between the gravitational potential due to a point mass and the electric potential due to a point charge.

similarity: ....

....

difference: ....

....

[2]

(b) An isolated uniform conducting sphere has mass M and charge Q.
The gravitational field strength at the surface of the sphere is g.
The electric field strength at the surface of the sphere is E.

(i) Show that

$$\frac{M}{Q} = \alpha \frac{g}{E}$$

where α is a constant.

[3]

(ii) Show that the numerical value of α is $1.35 \times 10^{20}\,\text{kg}^2\,\text{C}^{-2}$.

[1]

(c) Assume that the Earth is a uniform conducting sphere of mass 5.98×10^{24} kg.
The surface of the Earth carries a charge of -4.80×10^{5} C that is evenly distributed.

(i) Use the information in **(b)** to determine the electric field strength at the surface of the Earth. Give a unit with your answer.

electric field strength = ... unit .... [2]

(ii) State how the direction of the electric field at the surface of the Earth compares with the direction of the gravitational field.

.... [1]

[Total: 11]

53. O/N/23/41/Q5

(a) Define electric potential at a point.

....

....

.... [2]

(b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance x from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with x of the electric potential V due to the spheres at point P.

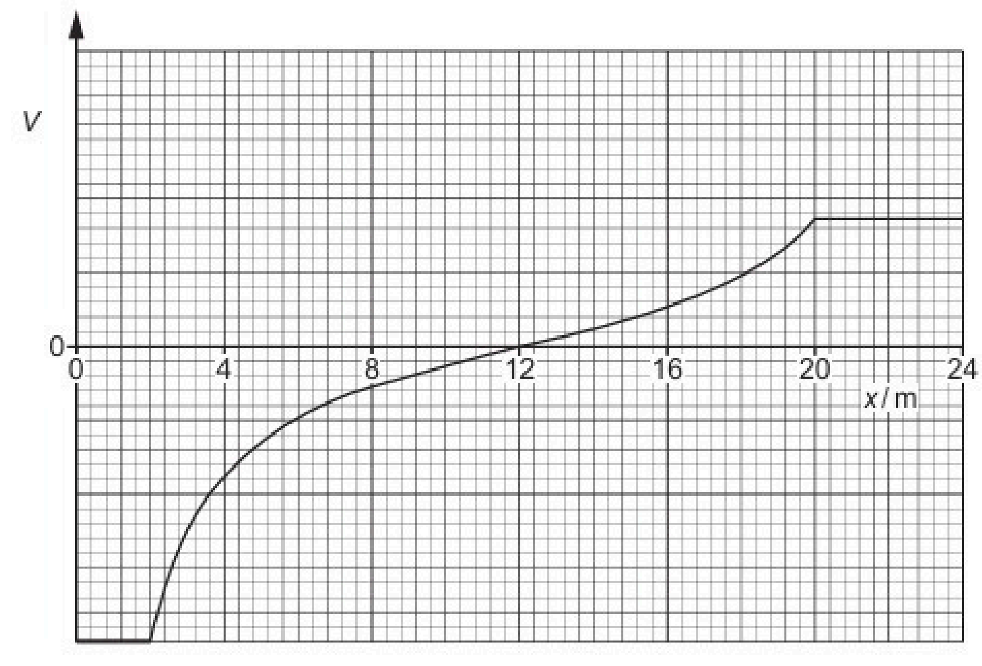


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1 ....

....

2 ....

....

3 ....

....

[3]

(c) A positively charged particle is placed at point P in **(b)**, such that $x = 12\,m$.
The particle is released.

Describe and explain the subsequent motion of the particle.

....

....

....

....

.... [3]

[Total: 8]

54. O/N/23/42/Q5

(a) State Coulomb's law.

....

....

.... [2]

(b) Two identical oil droplets are in a vacuum. The centres of the droplets are a distance of 3.8×10^{-6} m apart. The droplets have equal charge and exert an electric force on each other of magnitude 6.3×10^{-17} N.

Determine the magnitude of the charge on each droplet.

charge = ... C [2]

(c) One of the oil droplets in **(b)** is now placed between two horizontal metal plates, as shown in Fig. 5.1.

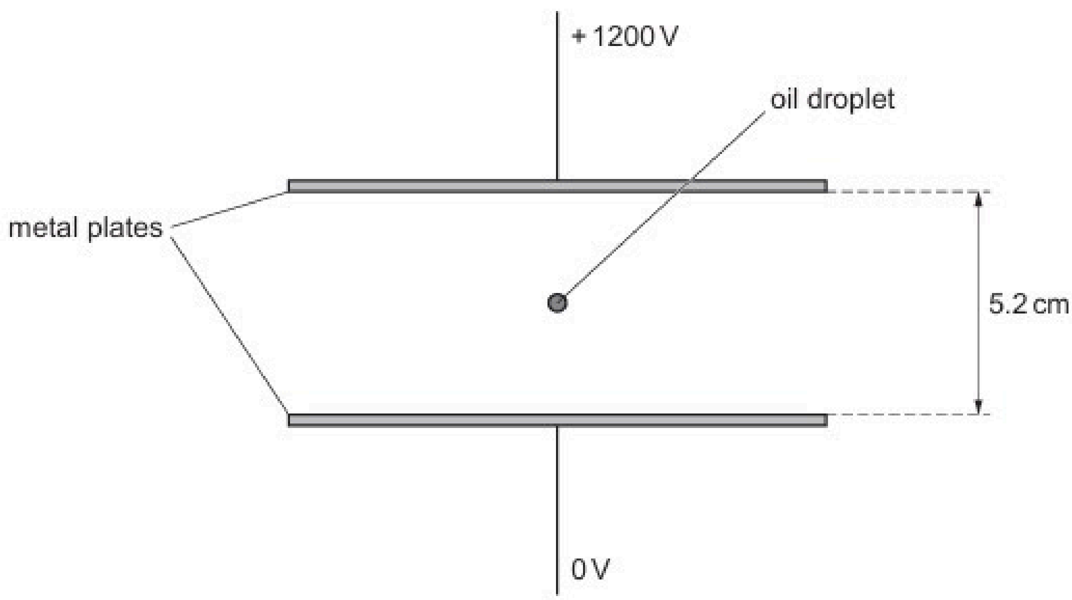


Fig. 5.1 (not to scale)

A potential difference (p.d.) of 1200 V is applied between the plates, with the top plate at the higher potential. The oil droplet is stationary and in equilibrium.

(i) State the sign of the charge on the oil droplet.

.... [1]

(ii) On Fig. 5.1, draw four lines to represent the electric field between the plates. [3]

(iii) The distance between the plates is 5.2 cm.

Determine the mass of the oil droplet.

mass = ... kg [3]

[Total: 11]

ANSWERS

1.

(a) field strength = potential gradient — M1
correct sign OR directions discussed — A1

(b) area is 21.2 $cm^2 \pm 0.4$ cm^2 — C2
(if outside ± 0.4 cm^2 but within ± 0.8 cm^2, allow 1 mark)
1.0 cm^2 represents ($1.0 \times 10^{-2} \times 2.5 \times 10^3$ =) 25 V — C1
potential difference = 530 V — A1

(c) $\frac{1}{2}mv^2 = qV$
$\frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 1.6 \times 10^{-19} \times 530$ — C1
$v = 1.37 \times 10^7$ ms^{-1} — A1

(d) **(i)** $d = 0$ — B1

(ii) acceleration decreases then increases — B1
some quantitative analysis (e.g. minimum at 4.0 cm) — B1
(any suggestion that acceleration becomes zero or that there is a deceleration scores 0/2)

2.

(a) **(i)** *either* lines directed away from sphere
or lines go from positive to negative
or line shows direction of force on positive charge M1
so positively charged A1

(ii) *either* all lines (appear to) radiate from centre
or all lines are normal to surface of sphere B1

(b) tangent to curve B1
in correct position and direction B1

(c) **(i)** $V = (0.76 \times 10^{-9}) / (4\pi \times 8.85 \times 10^{-12} \times 0.024)$ C1
$= 285$ V A1

(ii) negative charge is induced on (inside of) box M1
formula applies to isolated (point) charge
OR less work done moving test charge from infinity A1
so potential is lower A1

(d) *either* gravitational field is <u>always</u> attractive
or field lines must be directed towards both box and sphere B1

3.

(a) work done moving unit positive charge M1
from infinity to the point A1

(b) **(i)** $x = 18$ cm A1

(ii) $V_A + V_B = 0$ C1
$(3.6 \times 10^{-9}) / (4\pi\varepsilon_0 \times 18 \times 10^{-2}) + q / (4\pi\varepsilon_0 \times 12 \times 10^{-2}) = 0$ C1
$q = -2.4 \times 10^{-9}$ C A1
(use of $V_A = V_B$ giving 2.4×10^{-9} C scores one mark)

(c) field strength = (–) gradient of graph B1
force = charge × gradient / field strength *or force $\propto$ gradient* B1
force largest at $x = 27$ cm B1

4.

(a) work done per / on unit positive charge M1
moving charge from infinity to the point A1

(b) **(i)** α-particle and gold nucleus repel each other B1
<u>all</u> kinetic energy of α-particle converted into electric potential energy B1

(ii) **1** potential energy = $(79 \times 2 \times \{1.6 \times 10^{-19}\}^2) / (4\pi \times 8.85 \times 10^{-12} \times d)$ C1
kinetic energy = $4.8 \times 1.6 \times 10^{-13} = 7.68 \times 10^{-13}$ J C1
equating to give $d = 4.7 \times 10^{-14}$ m A1

(ii) **2** $F = Qq / 4\pi\varepsilon_0 d \times 1 / d = 7.68 \times 10^{-13} \times 1 / (4.7 \times 10^{-14})$ C1
$= 16$ N A1

5.

(a) ability to do work — B1
as a result of the position/shape, etc. of an object — B1

(b) (i) 1 $\Delta E_{gpe} = GMm / r$ — C1
$= (6.67 \times 10^{-11} \times \{2 \times 1.66 \times 10^{-27}\}^2) / (3.8 \times 10^{-15})$ — C1
$= 1.93 \times 10^{-49}$ J — A1

2 $\Delta E_{epe} = Qq / 4\pi\varepsilon_0 r$ — C1
$= (1.6 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times 3.8 \times 10^{-15})$ — C1
$= 6.06 \times 10^{-14}$ J — A1

(ii) idea that $2E_K = \Delta E_{epe} - \Delta E_{gpe}$ — B1
$E_K = 3.03 \times 10^{-14}$ J
$= (3.03 \times 10^{-14}) / 1.6 \times 10^{-13}$ — M1
$= 0.19$ MeV — A0

(iii) fusion may occur / may break into sub-nuclear particles — B1

6.

(a) force $= q_1 q_2 / 4\pi\varepsilon_0 x^2$ — C1
$= (6.4 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times \{12 \times 10^{-6}\}^2)$ — C1
$= 2.56 \times 10^{-17}$ N — A1

(b) potential at P is same as potential at Q — B1
work done $= q\Delta V$ — M1
$\Delta V = 0$ so zero work done — A0

(c) at midpoint, potential is $2 \times (6.4 \times 10^{-19}) / (4\pi\varepsilon_0 \times 6 \times 10^{-6})$ — C1
at P, potential is $(6.4 \times 10^{-19}) / (4\pi\varepsilon_0 \times 3 \times 10^{-6}) + (6.4 \times 10^{-19}) / (4\pi\varepsilon_0 \times 9 \times 10^{-6})$ — C1
change in potential $= (6.4 \times 10^{-19}) / (4\pi\varepsilon_0 \times 9 \times 10^{-6})$
energy $= 1.6 \times 10^{-19} \times (6.4 \times 10^{-19}) / (4\pi\varepsilon_0 \times 9 \times 10^{-6})$ — C1
$= 1.0 \times 10^{-22}$ J — A1

7.

(a) work done in bringing unit positive charge — M1
from infinity (to that point) — A1

(b) (i) field strength is potential gradient — B1

(ii) field strength proportional to force (on particle Q) — B1
potential gradient proportional to gradient of (potential energy) graph — B1
so force is proportional to the gradient of the graph — A0

(c) energy = $5.1 \times 1.6 \times 10^{-19}$ (J) C1
potential energy = $Q_1Q_2 / 4\pi\varepsilon_0 r$ C1
$5.1 \times 1.6 \times 10^{-19} = (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times r$ C1
$r = 2.8 \times 10^{-10}$ m A1

(d) **(i)** work is got out as x decreases M1
so opposite sign A1

(ii) energy would be doubled B1
gradient would be increased B1

8.

(a) region (of space) where a particle / body experiences a force B1

(b) similarity: e.g. force $\propto 1 / r^2$
potential $\propto 1 / r$ B1

difference: e.g. gravitation force (always) attractive B1
electric force attractive or repulsive B1

(c) *either* ratio is $Q_1Q_2 / 4\pi\varepsilon_0 m_1 m_2 G$ C1
$= (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times (1.67 \times 10^{-27})^2 \times 6.67 \times 10^{-11}$ C1
$= 1.2 \times 10^{36}$ A1
or $F_E = 2.30 \times 10^{-28} \times R^{-2}$ (C1)
$F_G = 1.86 \times 10^{-64} \times R^{-2}$ (C1)
$F_E / F_G = 1.2 \times 10^{36}$ (A1)

9.

(a) **(i)** zero field (strength) inside spheres B1

(ii) *either* field strength is zero
or the fields are in opposite directions M1
at a point between the spheres A1

(b) **(i)** field strength is (–) potential gradient *(not V/x)* B1

(ii) **1.** field strength has maximum value B1
at x = 11.4 cm B1

2. field strength is zero B1
either at x = 7.9 cm *(allow ±0.3 cm)*
or at 0 to 1.4 cm *or* 11.4 cm to 12 cm B1

10.

(a) force per unit positive charge acting on a stationary charge B1

(b) **(i)** $E = Q / 4\pi\varepsilon_0 r^2$ C1

$Q = 1.8 \times 10^4 \times 10^2 \times 4\pi \times 8.85 \times 10^{-12} \times (25 \times 10^{-2})^2$ M1

$Q = 1.25 \times 10^{-5}\,\text{C} = 12.5\,\mu\text{C}$ A0

(ii) $V = Q / 4\pi\varepsilon_0 r$

$= (1.25 \times 10^{-5}) / (4\pi \times 8.85 \times 10^{-12} \times 25 \times 10^{-2})$ C1

$= 4.5 \times 10^5\,\text{V}$ A1

(Do not allow use of V = Er unless explained)

11.

(a) **(i)** as *r* decreases, energy decreases/work got out (due to) M1

<u>attraction</u> so point mass is negatively charged A1

(ii) electric potential energy = charge × electric potential B1

electric field strength is potential gradient B1

field strength = gradient of potential energy graph/charge A0

(b) tangent drawn at (4.0, 14.5) B1

gradient = 3.6×10^{-24} A2

(*for < ±0.3 allow 2 marks, for < ±0.6 allow 1 mark*)

field strength= $(3.6 \times 10^{-24}) / (1.6 \times 10^{-19})$

$= 2.3 \times 10^{-5}\,\text{V m}^{-1}$ *(allow ecf from gradient value)* A1

(*one point solution for gradient leading to $2.3 \times 10^{-5}\,\text{Vm}^{-1}$ scores 1 mark only*)

12.

(a) **(i)** (tangent to line gives) direction of force on a (small test) mass B1

(ii) (tangent to line gives) direction of force on a (small test) charge M1

charge is positive A1

(b) similarity:
e.g. radial fields
lines normal to surface
greater separation of lines with increased distance from sphere
field strength $\propto$ 1 / (distance to centre of sphere)^2
(*allow any sensible answer*) B1

difference:
e.g. gravitational force (always) towards sphere B1
electric force direction depends on sign of charge on sphere / towards or away from sphere B1
e.g. gravitational field/force is attractive (B1)
electric field/force is attractive or repulsive (B1)
(*allow any sensible comparison*)

(c) gravitational force $= 1.67 \times 10^{-27} \times 9.81$
$= 1.6 \times 10^{-26}$ N A1
electric force $= 1.6 \times 10^{-19} \times 270 / (1.8 \times 10^{-2})$ C1
$= 2.4 \times 10^{-15}$ N A1
electric force very much greater than gravitational force B1

13.

(a) work done moving unit positive charge M1
from infinity (to the point) A1

(b) (gain in) kinetic energy = change in potential energy B1
$\frac{1}{2}mv^2 = qV$ leading to $v = (2Vq/m)^{\frac{1}{2}}$ B1

(c) *either* $(2.5 \times 10^5)^2 = 2 \times V \times 9.58 \times 10^7$ C1
$V =$ 330 V M1
this is less than 470 V and so 'no' A1

or $v = (2 \times 470 \times 9.58 \times 10^7)$ (C1)
$v = 3.0 \times 10^5$ m s^{-1} (M1)
this is greater than 2.5×10^5 m s^{-1} and so 'no' (A1)

or $(2.5 \times 10^5)^2 = 2 \times 470 \times (q/m)$ (C1)
$(q/m) = 6.6 \times 10^7$ C kg^{-1} (M1)
this is less than 9.58×10^7 C kg^{-1} and so 'no' (A1)

14.

(a) **(i)** force proportional to product of (two) charges and inversely proportional to square of separation — M1

reference to point charges — A1

(ii) $F = 2 \times (1.6 \times 10^{-19})^2 / \{4\pi \times 8.85 \times 10^{-12} \times (20 \times 10^{-6})^2\}$ — C1

$F = 1.15 \times 10^{-18}$ N — A1

(b) **(i)** force per unit charge — M1

on *either* a stationary charge

or a positive charge — A1

(ii) **1.** electric field is a vector quantity

electric fields are in opposite directions

charges repel

Any two of the above, 1 each — B2

2. graph: line always between given lines — M1

crosses x-axis between 11.0 μm and 12.3 μm — A1

reasonable shape for curve — A1

15.

(a) work done bringing unit positive charge — M1

from infinity (to the point) — A1

(b) **(i)** *either* both potentials are positive / same sign — M1

so same sign — A1

or gradients are positive & negative (so fields in opposite directions) — (M1)

so same sign — (A1)

(ii) the individual potentials are summed — B1

(iii) allow value of x between 10 nm and 13 nm — A1

(iv) $V = 0.43$ V (*allow 0.42 V → 0.44 V*) — M1

energy $= 2 \times 1.6 \times 10^{-19} \times 0.43$ — A1

$= 1.4 \times 10^{-19}$ J — A1

16.

(a) graph: straight line at constant potential = V_0 from $x = 0$ to $x = r$ — B1
curve with decreasing gradient — M1
passing through $(2r, 0.50V_0)$ and $(4r, 0.25V_0)$ — A1

(b) graph: straight line at $E = 0$ from $x = 0$ to $x = r$ — B1
curve with decreasing gradient from (r, E_0) — M1
passing through $(2r, \frac{1}{4}E_0)$ — A1
(*for 3rd mark line must be drawn to x = 4r and must not touch x-axis*)

17.

(a) **(i)** $F_E = Q_1Q_2/4\pi\varepsilon_0 r^2$ — C1
$= 8.99 \times 10^9 \times (1.6 \times 10^{-19})^2/(2.0 \times 10^{-15})^2$
$= 58$ N — A1

(ii) $F_G = Gm_1m_2/r^2$ — C1
$= 6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2/(2.0 \times 10^{-15})^2$
$= 4.7 \times 10^{-35}$ N — A1

(b) **(i)** force of <u>repulsion</u> (much) greater than force of <u>attraction</u> — B1
must be some other force of <u>attraction</u> — M1
to hold nucleus together — A1

(*Do not allow if $F_G > F_E$ in* ***(a)*** *or one of the forces not calculated in* ***(a)***)

(ii) outside nucleus there is repulsion between protons — B1
either attractive force must act only in nucleus
or if not short range, all nuclei would stick together — B1

18.

(a) work done in moving unit positive charge — M1
from infinity (to the point) — A1

(b) **(i)** inside the sphere, the potential would be constant — B1

(ii) for point charge, Vx is constant — B1
co-ordinates clear and determines two values of Vx at least 4 cm apart — M1
conclusion made clear — A1

(c) $q = 4\pi\varepsilon_0 Vx$
$q = 4\pi \times 8.85 \times 10^{-12} \times 180 \times 1.0 \times 10^{-2}$ — M1
$= 2.0 \times 10^{-10}$ C — A1

19.

(a) work done / energy in moving unit positive charge from infinity (to the point) — M1, A1

(b) **(i)** $V = q/4\pi\varepsilon_0 r$

at 16 kV, $q = 3.0 \times 10^{-8}$ C

$r = (3.0 \times 10^{-8})/(4\pi \times 8.85 \times 10^{-12} \times 16 \times 10^3)$ — C1

$= 1.69 \times 10^{-2}$ m (*allow 2 s.f.*) — A1

(*allow any answer which rounds to 1.7* $\times$ *10*$^{-2}$)

(ii) energy is / represented by area 'below' line — C1

energy $= \frac{1}{2}qV$

$= \frac{1}{2} \times 24 \times 10^3 \times 4.5 \times 10^{-8}$ — C1

$= 5.4 \times 10^{-4}$ J — A1

(c) $V = q/4\pi\varepsilon_0 r$ and $E = q/4\pi\varepsilon_0 r^2$ giving $Er = V$ — B1

$2.0 \times 10^6 \times 1.7 \times 10^{-2} = V$ — C1

$V = 3.4 \times 10^4$ V — A1

20.

(a) (magnitude of electric field strength is the potential gradient — B1

use of gradient at x = 4.0 cm — M1

gradient = 4.5×10^4 N C^{-1} (*allow* $\pm 0.3 \times 10^4$) — A1

or

$$V = \frac{Q}{4\pi\varepsilon_0 x} \text{ and } E = \frac{Q}{4\pi\varepsilon_0 x^2} \text{ leading to } E = \frac{V}{x}$$ (B1)

$E = 1.8 \times 10^3/0.04$ — (M1)

$= 4.5 \times 10^4$ N C^{-1} — (A1)

(b) **(i)** 3.6×10^3 V — A1

(ii) capacitance $= Q/V$ — C1

$= (8.0 \times 10^{-9})/(3.6 \times 10^3)$

$= 2.2 \times 10^{-12}$ F — A1

21.

(a)		work done bringing/moving per unit positive charge	M1
		from infinity (to the point)	A1
(b)	**(i)**	slope/gradient (of the line/graph/tangent)	B1
		*(allow dV/dx, but **not** $\Delta V/\Delta x$ or V/x)*	
		(allow potential gradient)	
		(negative sign not required)	
	(ii)	maximum at surface of sphere A or at $x = 0$ (cm)	B1
		zero at $x = 6$ (cm)	B1
		then increases but in opposite direction	B1
		(any mention of attraction max. 2/3)	
(c)	**(i)**	M shown between $x = 5.5$ cm and $x = 6.5$ cm	B1
	(ii)	**1.** $\Delta V = (570 - 230) = 340$ V *(allow 330 V to 340 V)*	A1
		2. $q(\Delta)V = ½mv^2$ **or** change/loss in PE = change/gain in KE **or** $\Delta E_K = \Delta E_P$	B1
		$4.8 \times 10^7 \times 340 = ½v^2$	C1
		$v^2 = 3.26 \times 10^{10}$	
		$v = 1.8 \times 10^5$ m s^{-1} *(not 1 s.f.)*	A1

22.

(a)	**(i)**	force proportional to the product of the two/point charges	B1
		and inversely proportional to the square of their separation	B1
	(ii)	**1.** force radially away from sphere/to right/to east	B1
		2. (maximum) at/on surface of sphere *or* $x = r$	B1
		3. $F \propto 1/x^2$ *or* $F = q_1q_2/(4\pi\varepsilon_0 x^2)$	C1
		ratio = 16	A1
(b)		$E = q/(4\pi\varepsilon_0 x^2)$ *or* $E \propto q$	C1
		maximum charge $= (2.0/1.5) \times 6.0 \times 10^{-7}$	C1
		$= 8.0 \times 10^{-7}$ C	
		additional charge = 2.0×10^{-7} C	A1

23.

(a) charges in metal do not move B1

no (resultant) force on charges so no (electric) field B1

(*allow 1/2 for "no field inside sphere"*)

(b) *either* average field strength $= \frac{1}{2}(28 + 54)\ \mathrm{N\,C^{-1}}$ C1

average force $= 8.5 \times 10^{-9} \times \frac{1}{2}(28 + 54)$ C1

$= 3.49 \times 10^{-7}\ \mathrm{N}$

change in potential energy $= 3.49 \times 10^{-7} \times 2.0 \times 10^{-2}$

$= 7.0 \times 10^{-9}\ \mathrm{J}$ (*allow 1 s.f.*) A1

(*allow range 54 ± 1*)

or (for a point charge) $V = Ex$ (C1)

$\Delta V = (54 \times 5.0 \times 10^{-2}) - (28 \times 7.0 \times 10^{-2})$ (C1)

change in potential energy $= 8.5 \times 10^{-9} \times (2.70 - 1.96)$

$= 6.3 \times 10^{-9}\ \mathrm{J}$ (*allow 1 s.f.*) (A1)

(*allow range 54 ± 1*)

or ΔV is area under curve (C1)

$\Delta V = 0.74\ \mathrm{V}$ (C1)

change in potential energy $= 8.5 \times 10^{-9} \times 0.74$

$= 6.3 \times 10^{-9}\ \mathrm{J}$ (*allow 1 s.f.*) (A1)

(*allow range 0.70 to 0.84*)

24.

(a) sketch: from $x = 0$ to $x = R$, potential is constant at V_S B1

smooth curve through (R, V_S) and $(2R, 0.5V_S)$ B1

smooth curve continues to $(3R, 0.33V_S)$ B1

(b) sketch: from $x = 0$ to $x = R$, field strength is zero B1

smooth curve through (R, E) and $(2R, 0.25E)$ B1

smooth curve continues to $(3R, 0.11E)$ B1

25.

(a) lines perpendicular to surface
or
lines are radial M1

lines <u>appear</u> to come from centre A1

(b) (i) $F_E = (1.6 \times 10^{-19})^2 / 4\pi\varepsilon_0 x^2$ C1

$F_G = G \times (1.67 \times 10^{-27})^2 / x^2$ C1

$F_E / F_G = (1.6 \times 10^{-19})^2 \times (8.99 \times 10^9) / [(1.67\times10^{-27})^2 \times (6.67\times10^{-11})]$
$= 1.2\ (1.24) \times 10^{36}$ A1

(ii) $F_E \gg F_G$ B1

26.

(a) in an electric field, charges (in a conductor) would move B1

no movement of charge so zero field strength
or
charge moves until $F = 0$ / $E = 0$ B1

or

charges in metal do not move (B1)
no (resultant) force on charges so no (electric) field (B1)

(b) at P, $E_A = (3.0 \times 10^{-12}) / [4\pi\varepsilon_0(5.0 \times 10^{-2})^2]$ (= 10.79 N C^{-1}) M1

at P, $E_B = (12 \times 10^{-12}) / [4\pi\varepsilon_0(10 \times 10^{-2})^2]$ (= 10.79 N C^{-1}) M1

or

$(3.0 \times 10^{-12}) / [4\pi\varepsilon_0(5.0 \times 10^{-2})^2] - (12 \times 10^{-12}) / [4\pi\varepsilon_0(10 \times 10^{-2})^2] = 0$
or
$(3.0 \times 10^{-12}) / [4\pi\varepsilon_0(5.0 \times 10^{-2})^2] = (12 \times 10^{-12}) / [4\pi\varepsilon_0(10 \times 10^{-2})^2]$ (M2)

fields due to charged spheres are (equal and) <u>opposite in direction</u>, so $E = 0$ A1

(c) potential $= 8.99 \times 10^9 \{(3.0 \times 10^{-12}) / (5.0 \times 10^{-2}) + (12 \times 10^{-12}) / (10 \times 10^{-2})\}$ C1

$= 1.62$ V A1

(d) $\frac{1}{2}mv^2 = qV$

$E_K = \frac{1}{2} \times 107 \times 1.66 \times 10^{-27} \times v^2$ C1

$qV = 47 \times 1.60 \times 10^{-19} \times 1.62$ C1

$v^2 = 1.37 \times 10^8$

$v = 1.2 \times 10^4\,\mathrm{m\,s^{-1}}$ A1

27.

(a) $E = 0$ *or* $E_A = (-)E_B$ (at $x = 11\,\mathrm{cm}$) B1

$Q_A / x^2 = Q_B / (20 - x)^2 = 11^2 / 9^2$ C1

Q_A / Q_B or ratio = 1.5 A1

or

$E \propto Q$ because r same *or* $E = Q / 4\pi\varepsilon_0 r^2$ and r same (B1)

$Q_A / Q_B = 48 / 32$ (C1)

Q_A / Q_B or ratio = 1.5 (A1)

(b) **(i)** for max. speed, $\Delta V = (0.76 - 0.18)\,\mathrm{V}$ or $\Delta V = 0.58\,\mathrm{V}$ C1

$q\Delta V = \frac{1}{2}mv^2$

$2 \times (1.60 \times 10^{-19}) \times 0.58 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$ C1

$v^2 = 5.59 \times 10^7$

$v = 7.5 \times 10^3\,\mathrm{m\,s^{-1}}$ A1

(ii) $\Delta V = 0.22\,\mathrm{V}$ C1

$2 \times (1.60 \times 10^{-19}) \times 0.22 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$

$v^2 = 2.12 \times 10^7$

$v = 4.6 \times 10^3\,\mathrm{m\,s^{-1}}$ A1

28.

5(a)	(loss in) kinetic energy of α-particle = $Qq / 4\pi\varepsilon_0 r$ **or** $7.7 \times 10^{-13} = Qq / 4\pi\varepsilon_0 r$	C1
	$7.7 \times 10^{-13} = 8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / r$	M1
	$r = 4.7 \times 10^{-14}$ m *r* is closest distance of approach so radius less than this	A1
5(b)	force = $Qq / 4\pi\varepsilon_0 r^2$ = 4u × *a*	C1
	$8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / (4.7 \times 10^{-14})^2 = 4 \times 1.66 \times 10^{-27} \times a$	C1
	$a = 2.5 \times 10^{27}$ m s^{-2}	A1
5(c)	so that single interactions between nucleus and α-particle can be studied **or** so that multiple deflections with nucleus do not occur	B1

29.

6(a)	force proportional to product of charges and inversely proportional to the square of the separation	M1
	reference to point charges	A1
6(b)(i)	(near to each sphere,) fields are in opposite directions **or** point (between spheres) where fields are equal and opposite **or** point (between spheres) where field strength is zero	M1
	so same (sign of charge)	A1
6(b)(ii)	(at *x* = 5.0 cm,) $E = 3.0 \times 10^3$ V m^{-1} **and** $a = qE / m$	C1
	$E = (1.60 \times 10^{-19} \times 3.0 \times 10^3) / (1.67 \times 10^{-27})$	C1
	$= 2.9 \times 10^{11}$ m s^{-2}	A1
6(c)	field strength or *E* is potential gradient **or** field strength is rate of change of (electric) potential	M1
	(field strength) maximum at *x* = 6 cm	A1

30.

6(a)	electric field lines are radial/normal to surface (of sphere)	B1
	electric field lines <u>appear</u> to originate from centre (of sphere)	B1
6(b)(i)	tangent drawn at *x* = 6.0 cm and gradient calculation attempted	C1
	$E = 9.0 \times 10^4$ N C^{-1} (*1 mark if in range ±1.2; 2 marks if in range ±0.6*)	A2
	or	
	correct pair of values of *V* and *x* read from curved part of graph and substituted into $V = q / 4\pi\varepsilon_0 x$	(C1)
	to give $q = 3.6 \times 10^{-8}$ C	(C1)
	(then $E = q / 4\pi\varepsilon_0 x^2$ and *x* = 6 cm gives) $E = 9.0 \times 10^4$ N C^{-1}	(A1)
	or	
	($E = q / 4\pi\varepsilon_0 x^2$ and $V = q / 4\pi\varepsilon_0 x$ and so) $E = V / x$	(C1)
	giving $E = 5.4 \times 10^3 / 0.060$	(C1)
	$= 9.0 \times 10^4$ N C^{-1}	(A1)
6(b)(ii)	(*R* =) 2.5 cm	B1
	potential inside a conductor is constant **or** field strength inside a conductor zero (so gradient is zero)	B1

31.

7(a)	work done per unit charge	**B1**
	(work done) moving positive charge from infinity (to the point)	**B1**
7(b)(i)	potential always same sign / potential is always positive so same sign of charge	**B1**
7(b)(ii)	1 *from x = 12 cm to x = 25 cm:* speed increases and *from x = 27 cm to x = 31 cm*: speed decreases	**B1**
	(*from x = 12 cm to x = 25 cm:* speed increases) at decreasing rate or (*from x = 27 cm to x = 31 cm*: speed decreases) at increasing rate	**B1**
	at x = 26 cm: speed maximum	**B1**
	at 32 cm: speed still decreasing	**B1**
	2 $q\,\Delta V = ½mv^2$ $3.2 \times 10^{-19} \times (2.14 - 1.43) \times 10^4 = ½ \times 6.6 \times 10^{-27} \times v^2$ $v^2 = 6.88 \times 10^{11}$	**C1**
	$v = \mathbf{8.3 \times 10^5}\ \mathrm{m\,s^{-1}}$ (8.30)	**A1**

32.

6(a)	force per unit charge	**B1**
6(b)	$E = Q/(4\pi\varepsilon_0 r^2)$	**C1**
	$2.0 \times 10^4 = Q/(4\pi \times 8.85 \times 10^{-12} \times 0.26^2)$ charge $= 1.5 \times 10^{-7}$ C	**A1**
6(c)	charge ($= Q\,[52/26]^2$) $= 4Q$	**C1**
	additional charge = 3Q	**A1**

33.

6(a)(i)	work done per unit charge	**B1**
	work done moving positive charge from infinity (to the point)	**B1**
6(a)(ii)	field strength = potential gradient	**M1**
	'–' sign included **or** directions discussed	**A1**
6(b)(i)	gain in kinetic energy (= loss in potential energy) = charge × p.d. **or** $qV = ½mv^2$	**M1**
	so v is independent of separation (because separation not in expressions)	**A1**
6(b)(ii)	(at x = 0.40 cm), potential = (–) 75 × 0.40 / 1.2 (= (–) 25 V)	**C1**
	$½mv^2 = qV$ $½ \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 25$	**C1**
	or	
	$a = Vq/dm$ **and** $v^2 = 2as$	**(C1)**
	$v^2 = (2 \times 75 \times 2 \times 1.60 \times 10^{-19} \times 0.40 \times 10^{-2}) / (1.2 \times 10^{-2} \times 4 \times 1.66 \times 10^{-27})$	**(C1)**
	$v = 4.9 \times 10^4\ \mathrm{m\,s^{-1}}$	**A1**

34.

6(a)(i)	work done per unit charge	**B1**
	work done moving positive charge from infinity (to the point)	**B1**
6(a)(ii)	field strength = potential gradient	**M1**
	negative sign included **or** directions discussed	**A1**
6(b)	horizontal straight lines, at non-zero potential, within the spheres	**B1**
	magnitude of potential greater at surface of sphere A than at surface of sphere B	**B1**
	concave curve between A and B, with a minimum nearer to B	**B1**
	lines show V positive all the way from 0 to D	**B1**

35.

5(a)	region where charge experiences an (electric) force	**B1**
5(b)	graph: field strength zero from $x = 0$ to $x = R$	**B1**
	curve with negative gradient, decreasing from $x = R$ to $x = 3R$	**B1**
	line passes through field strength E at $x = R$,	**B1**
	line passes through field strength $0.25E$ at $x = 2R$ and field strength $0.11E$ at $x = 3R$	**B1**
5(c)	field strength = $q / 4\pi\varepsilon_0 x^2$	**C1**
	$2.0 \times 10^6 = q / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.26^2)$	**C1**
	$q = 1.5 \times 10^{-5}$ C	**A1**

36.

5(a)	force per unit charge	**B1**
	(force on) positive charge	**B1**
5(b)(i)	field changes direction (between A and B)/field is zero at a point (between A and B)	**M1**
	so charges have same sign	**A1**
5(b)(ii)	Any one from: • field is (also) influenced by charge B • charge A is not isolated/is not the only charge present • field is due to two/both charges • field is the resultant of two fields	**B1**
5(b)(iii)	$E = Q / (4\pi\varepsilon_0 x^2)$	**C1**
	at $x = 10$ cm, $E_A = E_B$	**C1**
	$Q_A / 10^2 = Q_B / 5^2$ $Q_A / Q_B = 4.0$	**A1**

37.

6(a)	work done per unit charge	**B1**
	(work done) moving positive charge from infinity	**B1**
6(b)	straight line with non-zero gradient from $x = 0$ to $x = d$	**B1**
	line with gradient of constant sign and end-points between which $\Delta V = V_0$ and $\Delta x = d$	**B1**
	line passes through (d, 0) and (0, $+V_0$) with negative gradient throughout	**B1**
6(c)	V constant (and non-zero) from $0 \rightarrow R$ and from $(D - R) \rightarrow D$	**B1**
	equal (non-zero) values of (magnitude of) V at R and $(D - R)$.	**B1**
	curve (with a minimum) from R to $(D - R)$ with V always positive	**B1**
	minimum at mid-point of curve	**B1**

38.

6(a)	(E =) $Q / 4\pi\varepsilon_0 r^2$	**M1**
	where ε_0 is permittivity (of free space)	**A1**
6(b)(i)	field does not change direction/field does not become zero	**M1**
	so (charges have) opposite (sign)	**A1**
6(b)(ii)	minimum is at the midpoint (between the charges)	**M1**
	so (magnitudes are the) same	**A1**
6(c)	force = field strength × charge **and** force = mass × acceleration **or** acceleration is proportional to field strength	**B1**
	(from x = 3.0 cm) to x = 5.0 cm: acceleration decreases	**B1**
	at x = 5.0 cm: acceleration is a minimum	**B1**
	from x = 5.0 cm (to x = 7.0 cm): acceleration increases	**B1**

39.

9(a)	work done per unit charge	**B1**
	(work done) moving positive charge from infinity	**B1**
9(b)(i)	energy $= 4.8 \times 1.60 \times 10^{-13}$ $= 7.7 \times 10^{-13}$ J	**A1**
9(b)(ii)	$E_P = Qq / 4\pi\varepsilon_0 d$	**C1**
	$Q = 79e$ **and** $q = 2e$	**C1**
	$7.68 \times 10^{-13} = (79 \times 2 \times \{1.60 \times 10^{-19}\}^2 / (4\pi \times 8.85 \times 10^{-12} \times d)$	**C1**
	$d = 4.7 \times 10^{-14}$ m	**A1**
9(c)	(diameter must be) less than/equal to 10^{-13} or 10^{-14} m	**B1**

40.

6(a)	**2.0** cm	**B1**
6(b)	At 16 (cm) from A the electric fields are equal or $E_A = E_B$	**B1**
	$E = Q / 4\pi\varepsilon_o r^2$ $Q_A / (4\pi\varepsilon_o r_A^2) = Q_B / (4\pi\varepsilon_o r_B^2)$ $3.6 \times 10^{-9} / 0.16^2 = Q_B / 0.08^2$	**C1**
	Q_B = $\mathbf{9.0 \times 10^{-10}}$ C	**A1**
6(c)(i)	$V = Q / 4\pi\varepsilon_o r_A$ $V = 3.6 \times 10^{-9} / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.020)$	**C1**
	V = **1600** V	**A1**
6(c)(ii)	$C = Q / V$ $= 3.6 \times 10^{-9} / 1600$	**C1**
	$=$ $\mathbf{2.3 \times 10^{-12}}$ F	**A1**

41.

5(a)	similarity: both are radial **or** both have inverse square (variations)	**B1**
	difference: direction is always/only towards the mass **or** direction can be towards or away from charge	**B1**
5(b)	field strength = $Q / 4\pi\varepsilon_0 x^2$	**C1**
	$E = Q / 36\pi\varepsilon_0 R^2$	**A1**
5(c)(i)	fields (due to each sphere) are in same direction	**B1**
5(c)(ii)	charges on spheres attract/affect each other **or** charge distribution on each sphere distorted by the other sphere **or** charges on the surface of the spheres move	**B1**
	spheres are not point charges (at their centres)	**B1**

42.

7(a)	force per unit charge	**M1**
	(force on) positive charge	**A1**
7(b)(i)	no electric field inside a conductor	**B1**
	R = 4.5 cm	**A1**
7(b)(ii)	$E = Q / (4\pi\varepsilon_0 x^2)$	**C1**
	clear correct read-off of a pair of values of E and x	**C1**
	e.g. $Q = 18 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times (4.5 \times 10^{-2})^2$ $= 4.0 \times 10^{-7}$ C **or** 4.1×10^{-7} C	**A1**
7(c)	At 8.0 cm, $E = 5.75 \times 10^5$ V m^{-1}	**C1**
	$F = Eq$ **and** $a = F / m$	**C1**
	$F = (5.75 \times 10^5 \times 2 \times 1.6 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ $= 2.8 \times 10^{13}$ m s^{-2}	**A1**

43.

5(a)	work done per unit charge	**B1**
	(work done on charge) moving positive charge from infinity	**B1**
5(b)(i)	$(2.0 \times 10^{-9}) / 4\pi\varepsilon_0(4.0 \times 10^{-2}) + Q / 4\pi\varepsilon_0(8.0 \times 10^{-2}) = 0$	**C1**
	$Q = 4.0 \times 10^{-9}$ C	**A1**
	Q given with negative sign	**B1**
5(b)(ii)	change = 1200 V	**A1**
5(c)	$\frac{1}{2}mv^2 = qV$	**C1**
	$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 1200$	**C1**
	$v = 3.4 \times 10^5$ m s^{-1}	**A1**

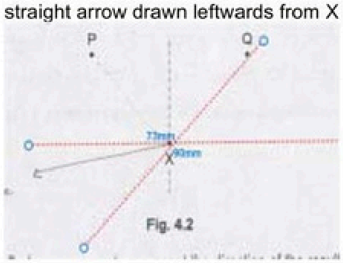
44.

5(a)(i)	region (of space)	**B1**
	where a particle experiences a force	**B1**
5(a)(ii)	similarity – any one point from: • both have an inverse square variation • both decrease with distance • both are radial	**B1**
	difference – any one point from: • gravitational field always towards (the mass) • electric field can be towards or away from (the charge)	**B1**
5(b)(i)	$E = Q / 4\pi\varepsilon_0 x^2$	**C1**
	$Q = 4\pi \times 8.85 \times 10^{-12} \times 84 \times 0.15^2$ $= 2.1 \times 10^{-10}$ C	**A1**
5(b)(ii)	$E = 84 \times (0.15 / 0.45)^2$ **or** $E = (2.1 \times 10^{-10}) / (4\pi \times 8.85 \times 10^{-12} \times 0.45^2)$	**C1**
	$E = 9.3$ V m^{-1}	**A1**
5(c)	line at $E = 0$ from $x = 0$ to $x = 0.15$ m	**B1**
	smooth curve with decreasing negative gradient throughout, from $x = 0.15$ m to $x = 0.45$ m, passing through (0.15, 84)	**B1**
	line passing through (0.45, 9.3)	**B1**

45.

6(a)	(both have) radial field lines	**B1**
6(b)(i)	**2.1** cm	**B1**
6(b)(ii)	$E = \dfrac{Q}{4\pi\varepsilon_o r^2}$ e.g. $r = 2.1$ cm, $E = 1.30 \times 10^5$ V m^{-1} $Q = 4\pi\varepsilon_o r^2 E$ $= 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021^2 \times 1.30 \times 10^5$	**C1**
	$= 6.4 \times 10^{-9} C$	**A1**
6(c)	$C = \dfrac{Q}{V}$ *either* $V = \dfrac{Q}{4\pi\varepsilon_o r}$ leading to $C = 4\pi\varepsilon_o r$	**C1**
	$C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021$	**C1**
	$(C =) 2.3 \times 10^{-12}$ **F**	**A1**
	or $V = \dfrac{Q}{4\pi\varepsilon_o r}$ $= \dfrac{6.4 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.021}$ $= 2740$V $C = \dfrac{6.4 \times 10^{-9}}{2740}$	**(C1)**
	$= 2.3 \times 10^{-12}$F	**(A1)**

46.

4(a)	direction of force	B1
	force on a positive charge	B1
4(b)(i)	$V = \dfrac{Q}{4\pi\varepsilon_o r}$ $\dfrac{4.0\times10^{-9}}{4\pi\varepsilon_o x} + \dfrac{-7.2\times10^{-9}}{4\pi\varepsilon_o(0.120 - x)} = 0$ $4(0.120 - x) = 7.2\,x$	C1
	x = 0.043 m	A1
4(b)(ii)	fields are in the same direction so no	B1
4(b)(iii)	straight arrow drawn leftwards from X in direction between extended line joining Q and X and the horizontal  Fig. 4.2	B1

47.

2(a)(i)	(vertically) downwards	B1
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	B1
	magnetic force perpendicular to velocity is the centripetal force **or** magnetic force perpendicular to velocity causes centripetal acceleration **or** acceleration perpendicular to velocity is centripetal (acceleration) **or** magnetic force does not change the speed of the sphere **or** magnetic force has constant magnitude	B1
2(b)	$mg = Eq$	C1
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8\ \text{N C}^{-1}$	A1
2(c)	centripetal force = magnetic force **or** $Bqv = mv^2 / r$	B1
	$B = mv / qr$	C1
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14\ \text{T}$	A1

48.

2(a)	(electric) force is (directly) proportional to product of charges	**B1**
	force (between point charges) is inversely proportional to the square of their separation	**B1**
2(b)(i)	(electric) force is perpendicular to velocity (of particles)	**B1**
	force (perpendicular to velocity) causes centripetal acceleration **or** force does not change the speed of the particles **or** force has constant magnitude	**B1**
2(b)(ii)	$F = e^2 / 4\pi\varepsilon_0 x^2$	**C1**
	$= (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2 \times 1.59 \times 10^{-10})^2]$ $= 2.28 \times 10^{-9}$ N	**A1**
2(b)(iii)	$F = mr\omega^2$ **and** $\omega = 2\pi / T$ **or** $F = mv^2 / r$ **<u>and</u>** $v = 2\pi r / T$	**C1**
	$F = 4\pi^2 mr / T^2$ $T = \sqrt{[4\pi^2 \times 9.11 \times 10^{-31} \times 1.59 \times 10^{-10} / (2.28 \times 10^{-9})]}$	**C1**
	$= 1.58 \times 10^{-15}$ s	**A1**
2(c)(i)	• electron and positron interact • positron is anti-particle of electron • (pair) annihilation occurs *Any two points, 1 mark each*	**B2**
	mass of the electron and positron converted into photon energy	**B1**
2(c)(ii)	PET scanning	**B1**

49.

4(a)	(field line indicates) direction of force	**B1**
	force on a positive charge	**B1**
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	**B1**
	five straight lines perpendicular to plates between the plates, uniformly spaced	**B1**
	downwards arrows on lines	**B1**
4(b)(ii)	$E = V / d$	**C1**
	$= 2400 / 0.046$ $= 5.2 \times 10^4$ N C^{-1}	**A1**
4(c)(i)	smooth curve in region of field and straight line outside field	**B1**
	direction of deflection shown as downwards in region of field	**B1**
4(c)(ii)	helium nucleus has double the charge but four times the mass	**B1**
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	**B1**
	final speed is lower (for helium)	**B1**

50.

5(a)	work done per unit charge	**B1**
	work done (on charge) in moving positive charge from infinity (to the point)	**B1**
5(b)(i)	radius = 0.060 m	**A1**
5(b)(ii)	$V = Q / 4\pi\varepsilon_0 x$ $Q = (-)\ 850 \times 4\pi \times 8.85 \times 10^{-12} \times 0.060$ **or** $Q = (-)\ 850 \times 0.060 / 8.99 \times 10^{9}$ (*any correct pair of V and x values from curve*)	**C1**
	$Q = -\ 5.7 \times 10^{-9}$ C	**A1**
5(c)(i)	$E_P = Q^2 / 4\pi\varepsilon_0 x$ $= (5.67 \times 10^{-9})^2 / (4\pi \times 8.85 \times 10^{-12} \times 0.46)$	**C1**
	$= 6.3 \times 10^{-7}$ J	**A1**
5(c)(ii)	• force is repulsive so spheres move apart • force in direction of motion so speed increases • potential energy converted to kinetic energy so speed increases • force decreases with distance so acceleration decreases • momentum is conserved (at zero) (and masses are equal) so velocities are always equal and opposite *Any three points, 1 mark each*	**B3**

51.

4(a)	(electric) force is (directly) proportional to product of charges	**B1**
	force (between point charges) is inversely proportional to the square of their separation	**B1**
4(b)(i)	arrows showing tension upwards in direction of string, electric force horizontally to the right and weight vertically downwards and all three labelled	**B1**
4(b)(ii)	$F_E = \dfrac{96 \times 10^{-9} \times 64 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080^2}$ $(= 8.63 \times 10^{-3}$ N)	**C1**
	either angle to vertical $= \sin^{-1} 0.080 / 1.2$ $(= 3.82°)$	**C1**
	weight $= F_E / \tan 3.82 = 8.63 \times 10^{-3} / \tan 3.82$ (= 0.129 N)	**C1**
	mass = 0.129 / 9.81 = 0.013 kg	**A1**
	or $T \sin\theta = mg$ **and** $T \cos\theta = F_E$ **or** $\tan\theta = mg / F_E$	**(C1)**
	$\tan\theta = 1.2 / 0.080$	**(C1)**
	$m = (1.2 \times 8.63 \times 10^{-3}) / (0.080 \times 9.81)$ = 0.013 kg	**(A1)**

4(b)(iii)	$E_p = \dfrac{Q_1Q_2}{4\pi\varepsilon_o r} = \dfrac{96\times10^{-9}\times64\times10^{-9}}{4\times\pi\times8.85\times10^{-12}\times0.080}$ $= 6.9 \times 10^{-4}$ J	A1
4(c)(i)	towards the top of the page / towards plate P	B1
4(c)(ii)	$F = QE$ **and** $E = V / d$	C1
	$F = 1.6 \times 10^{-19} \times 250 / 0.018$ $= 2.2 \times 10^{-15}$ N	A1
4(c)(iii)	**either** the force is not (always) perpendicular to the velocity **or** the force is always in the same direction	B1

52.

1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance *Any point, 1 mark*	B1
	difference: • gravitational potential is (always) negative • electric potential can be positive or negative *Any point, 1 mark*	B1
1(b)(i)	$g = GM / r^2$	M1
	$E = Q / 4\pi\varepsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M / Q = (1 / 4\pi G\varepsilon_0)\ (g / E)$	A1
1(b)(ii)	$\alpha = 1 / (4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20}$ (kg^2 C^{-2}) **or** $\alpha = (8.99 \times 10^9) / (6.67 \times 10^{-11}) = 1.35 \times 10^{20}$ (kg^2 C^{-2})	A1
1(c)(i)	$E = \alpha g Q / M$ $= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5) / (5.98 \times 10^{24})$	C1
	$= 106$ N C^{-1} **or** 106 V m^{-1}	A1
1(c)(ii)	same (direction)	B1

53.

5(a)	work done per unit charge	B1
	work (done on charge) moving positive charge from infinity (to the point)	B1
5(b)	*Any three points from*: *Up to 2 points from*: • radius of sphere X is 2.0 m • radius of sphere Y is 4.0 m • radius of Y is double the radius of X *Up to 2 points from*: • charge on X is negative • charge on Y is positive • spheres carry opposite charges *Up to 1 point from*: • magnitudes of charges on the spheres are equal	B3
5(c)	particle is attracted to X or repelled from Y **or** resultant force on particle is towards X / away from Y / to the left	B1
	particle accelerates towards X / away from Y / to the left	B1
	(magnitude of) acceleration of particle increases	B1

54.

5(a)	(electric) force is (directly) proportional to product of charges	B1
	(electric) force (between point charges) is inversely proportional to the square of their separation	B1
5(b)	$F = Q^2 / 4\pi\varepsilon_0 x^2$ $6.3 \times 10^{-17} = Q^2 / [4\pi \times 8.85 \times 10^{-12} \times (3.8 \times 10^{-6})^2]$	C1
	charge = 3.2×10^{-19} C	A1
5(c)(i)	negative	B1
5(c)(ii)	four straight lines perpendicular to the plates, starting on one plate and finishing on the other	B1
	lines equally spaced	B1
	arrows indicating direction downwards	B1
5(c)(iii)	$E = V / d$	C1
	$mg = EQ$	C1
	mass $= (1200 \times 3.2 \times 10^{-19}) / (9.81 \times 0.052)$ $= 7.5 \times 10^{-16}$ kg	A1